Chapter K
Design of HSS and Box Member Connections

Examples K.1 through K.6 illustrate common beam to column shear connections that have been adapted for use with HSS columns. Example K.7 illustrates a through-plate shear connection, which is unique to HSS columns. Calculations for transverse and longitudinal forces applied to HSS are illustrated in Examples K.8 and K.9. An example of an HSS truss connection is given in Example K.10. Examples of HSS cap plate, base plate and end plate connections are given in Examples K.11 through K.13.

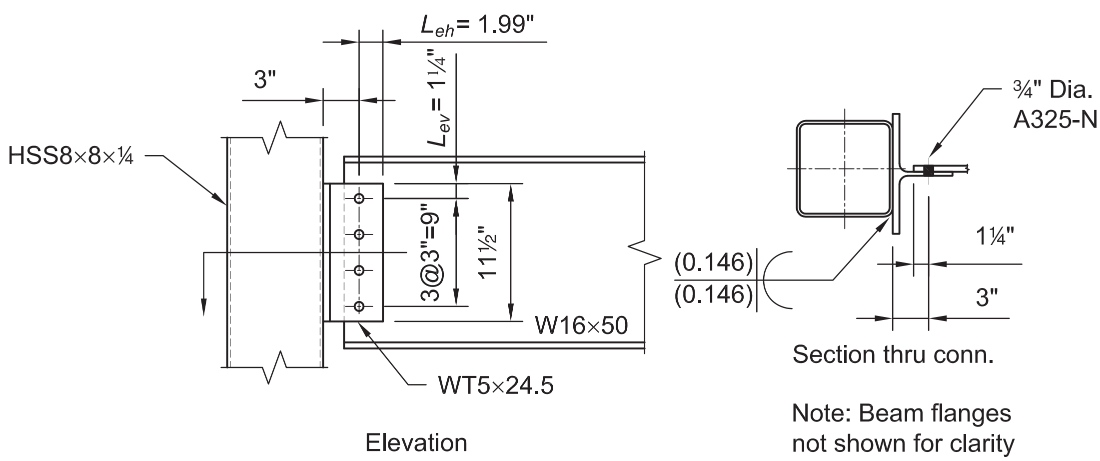
EXAMPLE K.1 WELDED/BOLTED WIDE TEE CONNECTION TO AN HSS COLUMN

Given:

Design a connection between an ASTM A992 W16×50 beam and an ASTM A500 Grade B HSS8×8×¼ column using an ASTM A992 WT5×24.5. Use ¾-in.-diameter ASTM A325-N bolts in standard holes with a bolt spacing, *s*, of 3 in., vertical edge distance L_{ev} of 1¼ in. and 3 in. from the weld line to the bolt line. Design as a flexible connection for the following vertical shear loads:

P_D = 6.20 kips
P_L = 18.5 kips

Note: A tee with a flange width wider than 8 in. was selected to provide sufficient surface for flare bevel groove welds on both sides of the column, because the tee will be slightly offset from the column centerline.



Solution:

From AISC *Manual* Table 2-4, the material properties are as follows:

Beam
ASTM A992
F_y = 50 ksi
F_u = 65 ksi

Tee
ASTM A992
F_y = 50 ksi
F_u = 65 ksi

Column
ASTM A500 Grade B
F_y = 46 ksi
F_u = 58 ksi

From AISC *Manual* Tables 1-1, 1-8 and 1-12, the geometric properties are as follows:

W16×50
t_w = 0.380 in.
d = 16.3 in.
t_f = 0.630 in.
T = 13⅝ in.

WT5×24.5
t_s = t_w = 0.340 in.
d = 4.99 in.
t_f = 0.560 in.
b_f = 10.0 in.
k_1 = 13⁄16 in.

HSS8×8×¼
t = 0.233 in.
B = 8.00 in.

From Chapter 2 of ASCE/SEI 7, the required strength is:

LRFD	ASD
P_u = 1.2(6.20 kips) + 1.6(18.5 kips) = 37.0 kips	P_a = 6.20 kips + 18.5 kips = 24.7 kips

Calculate the available strength assuming the connection is flexible.

Required Number of Bolts

The required number of bolts will ultimately be determined using the coefficient, C, from AISC *Manual* Table 7-6. First, the available strength per bolt must be determined.

Determine the available shear strength of a single bolt.

LRFD	ASD
ϕr_n = 17.9 kips from AISC *Manual* Table 7-1	$\dfrac{r_n}{\Omega}$ = 11.9 kips from AISC *Manual* Table 7-1

Determine single bolt bearing strength based on edge distance.

LRFD	ASD
L_{ev} = 1¼ in. ≥ 1 in. from AISC *Specification* Table J3.4 From AISC *Manual* Table 7-5, ϕr_n = 49.4 kips/in.(0.340 in.) = 16.8 kips	L_{ev} = 1¼ in. ≥ 1 in. from AISC *Specification* Table J3.4 From AISC *Manual* Table 7-5, $\dfrac{r_n}{\Omega}$ = 32.9 kips/in.(0.340 in.) = 11.2 kips

Determine single bolt bearing capacity based on spacing and AISC *Specification* Section J3.3.

LRFD	ASD
s = 3.00 in. ≥ 2⅔(¾ in.) = 2.00 in. From AISC *Manual* Table 7-4, ϕr_n = 87.8 kips/in.(0.340 in.) = 29.9 kips	s = 3.00 in. ≥ 2⅔(¾ in.) = 2.00 in. From AISC *Manual* Table 7-4, $\dfrac{r_n}{\Omega}$ = 58.5 kips/in.(0.340 in.) = 19.9 kips

Bolt bearing strength based on edge distance controls over available shear strength of the bolt.

Determine the coefficient for the eccentrically loaded bolt group.

LRFD	ASD
$C_{min} = \dfrac{P_u}{\phi r_n}$ $= \dfrac{37.0\text{ kips}}{16.8\text{ kips}}$ $= 2.20$ Using e = 3.00 in. and s = 3.00 in., determine C from AISC *Manual* Table 7-6. Try four rows of bolts, $C = 2.81 > 2.20$ **o.k.**	$C_{min} = \dfrac{P_a}{r_n / \Omega}$ $= \dfrac{24.7\text{ kips}}{11.2\text{ kips}}$ $= 2.21$ Using e = 3.00 in. and s = 3.00 in., determine C from AISC *Manual* Table 7-6. Try four rows of bolts, $C = 2.81 > 2.21$ **o.k.**

WT *Stem Thickness and Length*

AISC *Manual* Part 9 stipulates a maximum tee stem thickness that should be provided for rotational ductility as follows:

$$t_{s\,max} = \frac{d_b}{2} + \tfrac{1}{16}\text{ in.} \qquad \text{(\textit{Manual} Eq. 9-38)}$$

$$= \frac{(\tfrac{3}{4}\text{ in.})}{2} + \tfrac{1}{16}\text{ in.}$$

$$= 0.438\text{ in.} > 0.340\text{ in.} \quad \textbf{o.k.}$$

Note: The beam web thickness is greater than the WT stem thickness. If the beam web were thinner than the WT stem, this check could be satisfied by checking the thickness of the beam web.

Determine the length of the WT required as follows:

A W16×50 has a T-dimension of 13⅝ in.

L_{min} = T/2 from AISC *Manual* Part 10
= (13⅝ in.)/2
= 6.81 in.

Determine WT length required for bolt spacing and edge distances.

$L = 3(3.00 \text{ in.}) + 2(1\tfrac{1}{4} \text{ in.})$
$= 11.5 \text{ in.} < T = 13\tfrac{5}{8} \text{ in.}$ **o.k.**

Try $L = 11.5$ in.

Stem Shear Yielding Strength

Determine the available shear strength of the tee stem based on the limit state of shear yielding from AISC *Specification* Section J4.2.

$$R_n = 0.6F_y A_{gv} \qquad (Spec.\ \text{Eq. J4-3})$$
$$= 0.6(50 \text{ ksi})(11.5 \text{ in.})(0.340 \text{ in.})$$
$$= 117 \text{ kips}$$

LRFD	ASD
$\phi R_n = 1.00(117 \text{ kips})$ $= 117 \text{ kips}$ 117 kips > 37.0 kips **o.k.**	$\dfrac{R_n}{\Omega} = \dfrac{117 \text{ kips}}{1.50}$ $= 78.0 \text{ kips}$ 78.0 kips > 24.7 kips **o.k.**

Because of the geometry of the WT and because the WT flange is thicker than the stem and carries only half of the beam reaction, flexural yielding and shear yielding of the flange are not critical limit states.

Stem Shear Rupture Strength

Determine the available shear strength of the tee stem based on the limit state of shear rupture from AISC *Specification* Section J4.2.

$$R_n = 0.6F_u A_{nv} \qquad (Spec.\ \text{Eq. J4-4})$$
$$= 0.6F_u\left[L - n(d_h + \tfrac{1}{16} \text{ in.})\right](t_s)$$
$$= 0.6(65 \text{ ksi})[11.5 \text{ in.} - 4(\tfrac{13}{16} \text{ in.} + \tfrac{1}{16} \text{ in.})](0.340 \text{ in.})$$
$$= 106 \text{ kips}$$

LRFD	ASD
$\phi R_n = 0.75(106 \text{ kips})$ $= 79.5 \text{ kips}$ 79.5 kips > 37.0 kips **o.k.**	$\dfrac{R_n}{\Omega} = \dfrac{106 \text{ kips}}{2.00}$ $= 53.0 \text{ kips}$ 53.0 kips > 24.7 kips **o.k.**

Stem Block Shear Rupture Strength

Determine the available strength for the limit state of block shear rupture from AISC *Specification* Section J4.3.

For this case $U_{bs} = 1.0$.

Use AISC *Manual* Tables 9-3a, 9-3b and 9-3c. Assume $L_{eh} = 1.99 \text{ in.} \approx 2.00 \text{ in.}$

LRFD	ASD
$\dfrac{\phi F_u A_{nt}}{t} = 76.2$ kips/in.	$\dfrac{F_u A_{nt}}{t\Omega} = 50.8$ kips/in.
$\dfrac{\phi 0.60 F_y A_{gv}}{t} = 231$ kips/in.	$\dfrac{0.60 F_y A_{gv}}{t\Omega} = 154$ kips/in.
$\dfrac{\phi 0.60 F_u A_{nv}}{t} = 210$ kips/in.	$\dfrac{0.60 F_u A_{nv}}{t\Omega} = 140$ kips/in.
$\phi R_n = \phi 0.60 F_u A_{nv} + \phi U_{bs} F_u A_{nt}$ $\leq \phi 0.60 F_y A_{gv} + \phi U_{bs} F_u A_{nt}$ (from *Spec*. Eq. J4-5)	$\dfrac{R_n}{\Omega} = \dfrac{0.60 F_u A_{nv}}{\Omega} + \dfrac{U_{bs} F_u A_{nt}}{\Omega}$ $\leq \dfrac{0.60 F_y A_{gv}}{\Omega} + \dfrac{U_{bs} F_u A_{nt}}{\Omega}$ (from *Spec*. Eq. J4-5)
$\phi R_n = 0.340$ in.(210 kips/in.+76.2 kips/in.) ≤ 0.340 in.(231kips/in.+76.2 kips/in.) $= 97.3$ kips ≤ 104 kips	$\dfrac{R_n}{\Omega} = 0.340$ in.(140 kips/in. + 50.8 kips/in.) ≤ 0.340 in.(154 kips/in. + 50.8 kips/in.) $= 64.9$ kips ≤ 69.6 kips
97.3 kips > 37.0 kips **o.k.**	64.9 kips > 24.7 kips **o.k.**

Stem Flexural Strength

The required flexural strength for the tee stem is,

LRFD	ASD
$M_u = P_u e$ $= 37.0$ kips(3.00 in.) $= 111$ kip-in.	$M_a = P_a e$ $= 24.7$ kips(3.00 in.) $= 74.1$ kip-in.

The tee stem available flexural strength due to yielding is determined as follows, from AISC *Specification* Section F11.1. The stem, in this case, is treated as a rectangular bar.

$$Z = \frac{t_s d^2}{4} = \frac{0.340\text{ in.}(11.5\text{ in.})^2}{4} = 11.2\text{ in.}^3$$

$$S = \frac{t_s d^2}{6} = \frac{0.340\text{ in.}(11.5\text{ in.})^2}{6} = 7.49\text{ in.}^3$$

$$M_n = M_p = F_y Z \leq 1.6 M_y \qquad (\textit{Spec}.\text{ Eq. F11-1})$$
$$= 50\text{ ksi}\left(11.2\text{ in.}^3\right) \leq 1.6(50\text{ ksi})\left(7.49\text{ in.}^3\right)$$
$$= 560\text{ kip-in.} \leq 599\text{ kip-in.}$$

Therefore, use M_n = 560 kip-in.

Note: The 1.6 limit will never control for a plate, because the shape factor for a plate is 1.5.

The tee stem available flexural yielding strength is:

LRFD	ASD
$\phi M_n = 0.90(560 \text{ kip-in.})$ $= 504 \text{ kip-in.} > 111 \text{ kip-in.}$ **o.k.**	$\dfrac{M_n}{\Omega} = \dfrac{560 \text{ kip-in.}}{1.67}$ $= 335 \text{ kip-in.} > 74.1 \text{ kip-in.}$ **o.k.**

The tee stem available flexural strength due to lateral-torsional buckling is determined from Section F11.2.

$$\frac{L_b d}{t_s^2} = \frac{(3.00\text{ in.})(11.5\text{ in.})}{(0.340\text{ in.})^2}$$
$$= 298$$

$$\frac{0.08E}{F_y} = \frac{0.08(29{,}000\text{ ksi})}{50\text{ ksi}}$$
$$= 46.4$$

$$\frac{1.9E}{F_y} = \frac{1.9(29{,}000\text{ ksi})}{50\text{ ksi}}$$
$$= 1{,}102$$

Because 46.4 < 298 < 1,102, Equation F11-2 is applicable.

$$M_n = C_b\left[1.52 - 0.274\left(\frac{L_b d}{t^2}\right)\frac{F_y}{E}\right]M_y \le M_p \qquad (\textit{Spec}.\text{ Eq. F11-2})$$
$$= 1.00\left[1.52 - 0.274(298)\frac{50\text{ ksi}}{29{,}000\text{ ksi}}\right](50\text{ ksi})(7.49\text{ ksi}) \le (50\text{ ksi})(11.2\text{ in.}^3)$$
$$= 517\text{ kip-in.} \le 560\text{ kip-in.}$$

LRFD	ASD
$\phi = 0.90$ $\phi M_n = 0.90(517 \text{ kip-in.})$ $= 465 \text{ kip-in.} > 111 \text{ kip-in.}$ **o.k.**	$\Omega = 1.67$ $\dfrac{M_n}{\Omega_b} = \dfrac{517\text{ kip-in.}}{1.67}$ $= 310\text{ kip-in.} > 74.1\text{ kip-in.}$ **o.k.**

The tee stem available flexural rupture strength is determined from Part 9 of the AISC *Manual* as follows:

$$Z_{net} = \frac{td^2}{4} - 2t\left(d_h + \tfrac{1}{16}\text{ in.}\right)\left(1.5\text{ in.} + 4.5\text{ in.}\right)$$
$$= \frac{0.340\text{ in.}(11.5\text{ in.})^2}{4} - 2(0.340\text{ in.})\left(\tfrac{13}{16}\text{ in.} + \tfrac{1}{16}\text{ in.}\right)(1.5\text{ in.} + 4.5\text{ in.})$$
$$= 7.67\text{ in.}^3$$

$$M_n = F_u Z_{net} \qquad \text{(\textit{Manual} Eq. 9-4)}$$
$$= 65 \text{ ksi}\left(7.67 \text{ in.}^3\right)$$
$$= 499 \text{ kip-in.}$$

LRFD	ASD
$\phi M_n = 0.75(499 \text{ kip-in.})$ $= 374 \text{ kip-in.} > 111 \text{ kip-in.}$ **o.k.**	$\dfrac{M_n}{\Omega} = \dfrac{499 \text{ kip-in.}}{2.00}$ $= 250 \text{ kip-in.} > 74.1 \text{ kip-in.}$ **o.k.**

Beam Web Bearing

$$t_w > t_s$$
$$0.380 \text{ in.} > 0.340 \text{ in.}$$

Beam web is satisfactory for bearing by comparison with the WT.

Weld Size

Because the flange width of the WT is larger than the width of the HSS, a flare bevel groove weld is required. Taking the outside radius as 2(0.233 in.) = 0.466 in. and using AISC *Specification* Table J2.2, the effective throat thickness of the flare bevel weld is E = 5⁄16(0.466 in.) = 0.146 in.

Using AISC *Specification* Table J2.3, the minimum effective throat thickness of the flare bevel weld, based on the 0.233 in. thickness of the HSS column, is 1⁄8 in.

$$E = 0.146 \text{ in.} > \tfrac{1}{8} \text{ in.}$$

The equivalent fillet weld that provides the same throat dimension is:

$$\left(\frac{D}{16}\right)\left(\frac{1}{\sqrt{2}}\right) = 0.146$$
$$D = 16\sqrt{2}\left(0.146\right)$$
$$= 3.30 \text{ sixteenths of an inch}$$

The equivalent fillet weld size is used in the following calculations.

Weld Ductility

Let $b_f = B = 8.00$ in.

$$b = \frac{b_f - 2k_1}{2}$$
$$= \frac{8.00\text{ in.} - 2\left(^{13}\!/_{16}\text{ in.}\right)}{2}$$
$$= 3.19\text{ in.}$$

$$w_{min} = 0.0155\frac{F_y t_f^{\,2}}{b}\left(\frac{b^2}{L^2} + 2\right) \le \left(^5\!/_8\right)t_s \qquad (\textit{Manual}\text{ Eq. 9-36})$$
$$= 0.0155\frac{50\text{ ksi}\left(0.560\text{ in.}\right)^2}{3.19\text{ in.}}\left[\frac{\left(3.19\text{ in.}\right)^2}{\left(11.5\text{ in.}\right)^2} + 2\right] \le \left(^5\!/_8\right)\left(0.340\text{ in.}\right)$$
$$= 0.158\text{ in.} \le 0.213\text{ in.}$$

0.158 in. = 2.53 sixteenths of an inch

$D_{min} = 2.53 < 3.30$ sixteenths of an inch **o.k.**

Nominal Weld Shear Strength

The load is assumed to act concentrically with the weld group (flexible connection).

$a = 0$, therefore, $C = 3.71$ from AISC *Manual* Table 8-4

$$R_n = CC_1Dl$$
$$= 3.71(1.00)(3.30\text{ sixteenths of an inch})(11.5\text{ in.})$$
$$= 141\text{ kips}$$

Shear Rupture of the HSS at the Weld

$$t_{min} = \frac{3.09D}{F_u} \qquad (\textit{Manual}\text{ Eq. 9-2})$$
$$= \frac{3.09\left(3.30\text{ sixteenths}\right)}{58\text{ ksi}}$$
$$= 0.176\text{ in.} < 0.233\text{ in.}$$

By inspection, shear rupture of the WT flange at the welds will not control.

Therefore, the weld controls.

From AISC *Specification* Section J2.4, the available weld strength is:

LRFD	ASD
$\phi = 0.75$ $\phi R_n = 0.75(141\text{ kips})$ $= 106\text{ kips} > 37.0\text{ kips}$ **o.k.**	$\Omega = 2.00$ $\dfrac{R_n}{\Omega} = \dfrac{141\text{ kips}}{2.00}$ $= 70.5\text{ kips} > 24.7\text{ kips}$ **o.k.**

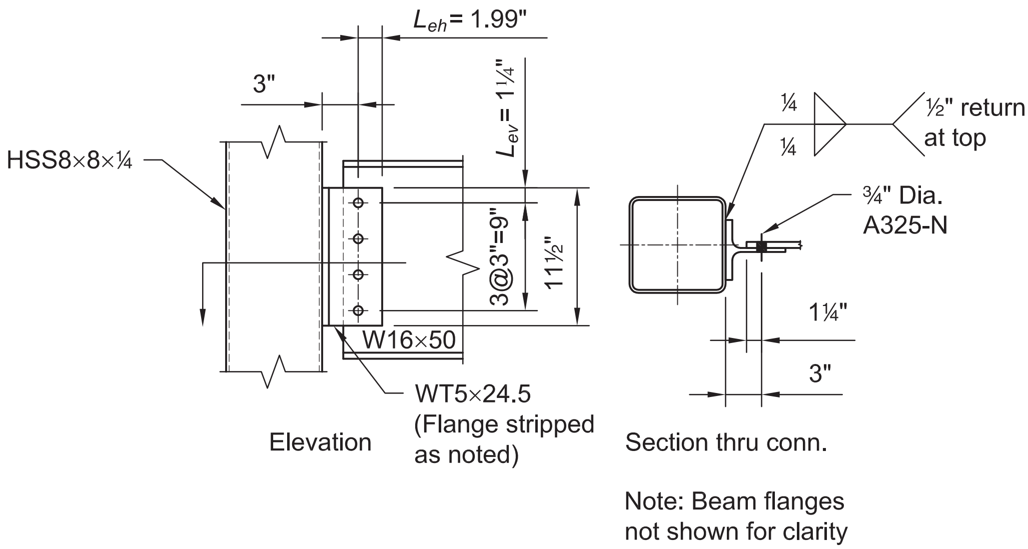
EXAMPLE K.2 WELDED/BOLTED NARROW TEE CONNECTION TO AN HSS COLUMN

Given:

Design a connection for an ASTM A992 W16×50 beam to an ASTM A500 Grade B HSS8×8×¼ column using a tee with fillet welds against the flat width of the HSS. Use ¾-in.-diameter A325-N bolts in standard holes with a bolt spacing, s, of 3.00 in., vertical edge distance L_{ev} of 1¼ in. and 3.00 in. from the weld line to the center of the bolt line. Use 70-ksi electrodes. Assume that, for architectural purposes, the flanges of the WT from the previous example have been stripped down to a width of 5 in. Design as a flexible connection for the following vertical shear loads:

P_D = 6.20 kips
P_L = 18.5 kips

Note: This is the same problem as Example K.1 with the exception that a narrow tee will be selected which will permit fillet welds on the flat of the column. The beam will still be centered on the column centerline; therefore, the tee will be slightly offset.



Solution:

From AISC *Manual* Table 2-4, the material properties are as follows:

Beam
ASTM A992
F_y = 50 ksi
F_u = 65 ksi

Tee
ASTM A992
F_y = 50 ksi
F_u = 65 ksi

Column
ASTM A500 Grade B
F_y = 46 ksi
F_u = 58 ksi

From AISC *Manual* Tables 1-1, 1-8 and 1-12, the geometric properties are as follows:

W16×50
t_w = 0.380 in.
d = 16.3 in.
t_f = 0.630 in.

HSS8×8×¼
t = 0.233 in.
B = 8.00 in.

WT5×24.5
$t_s = t_w$ = 0.340 in.
d = 4.99 in.
t_f = 0.560 in.
k_1 = 13⁄16 in.

From Chapter 2 of ASCE/SEI 7, the required strength is:

LRFD	ASD
P_u = 1.2(6.20 kips) + 1.6(18.5 kips) = 37.0 kips	P_a = 6.20 kips + 18.5 kips = 24.7 kips

The WT stem thickness, WT length, WT stem strength and beam web bearing strength are verified in Example K.1. The required number of bolts is also determined in Example K.1.

Maximum WT Flange Width

Assume ¼-in. welds and HSS corner radius equal to 2.25 times the nominal thickness 2.25(¼ in.) = 9⁄16 in.

The recommended minimum shelf dimension for ¼-in. fillet welds from AISC *Manual* Figure 8-11 is ½ in.

Connection offset:

$$\frac{0.380\text{ in.}}{2}+\frac{0.340\text{ in.}}{2}=0.360\text{ in.}$$

$$b_f \le 8.00\text{ in.} - 2\left(\tfrac{9}{16}\text{ in.}\right) - 2\left(\tfrac{1}{2}\text{ in.}\right) - 2\left(0.360\text{ in.}\right)$$

5.00 in. ≤ 5.16 in. **o.k.**

Minimum Fillet Weld Size

From AISC *Specification* Table J2.4, the minimum fillet weld size = ⅛ in. (D = 2) for welding to 0.233-in. material.

Weld Ductility

As defined in Figure 9-5 of the AISC *Manual* Part 9,

$$b=\frac{b_f-2k_1}{2}$$

$$= \frac{5.00 \text{ in.} - 2\left(1\tfrac{3}{16} \text{ in.}\right)}{2}$$
$$= 1.69 \text{ in.}$$

$$w_{min} = 0.0155 \frac{F_y t_f^{\,2}}{b}\left(\frac{b^2}{L^2} + 2\right) \leq \left(\tfrac{5}{8}\right) t_s \qquad \text{(\textit{Manual} Eq. 9-36)}$$

$$= 0.0155 \frac{50 \text{ ksi}\left(0.560 \text{ in.}\right)^2}{1.69 \text{ in.}}\left[\frac{\left(1.69 \text{ in.}\right)^2}{\left(11.5 \text{ in.}\right)^2} + 2\right] \leq \left(\tfrac{5}{8}\right)\left(0.340 \text{ in.}\right)$$
$$= 0.291 \text{ in.} \leq 0.213 \text{ in.}, \text{ use } 0.213 \text{ in.}$$

D_{min} = 0.213 in.(16)
= 3.41 sixteenths of an inch.

Try a ¼-in. fillet weld as a practical minimum, which is less than the maximum permitted weld size of t_f – 1⁄16 in. = 0.560 in. – 1⁄16 in. = 0.498 in. Provide ½-in. return welds at the top of the WT to meet the criteria listed in AISC *Specification* Section J2.2b.

Minimum HSS Wall Thickness to Match Weld Strength

$$t_{min} = \frac{3.09D}{F_u} \qquad \text{(\textit{Manual} Eq. 9-2)}$$
$$= \frac{3.09(4)}{58 \text{ ksi}}$$
$$= 0.213 \text{ in.} < 0.233 \text{ in.}$$

By inspection, shear rupture of the flange of the WT at the welds will not control.

Therefore, the weld controls.

Available Weld Shear Strength

The load is assumed to act concentrically with the weld group (flexible connection).

a = 0, therefore, C = 3.71 from AISC *Manual* Table 8-4

$R_n = CC_1Dl$
= 3.71(1.00)(4 sixteenths of an inch)(11.5 in.)
= 171 kips

From AISC *Specification* Section J2.4, the available fillet weld shear strength is:

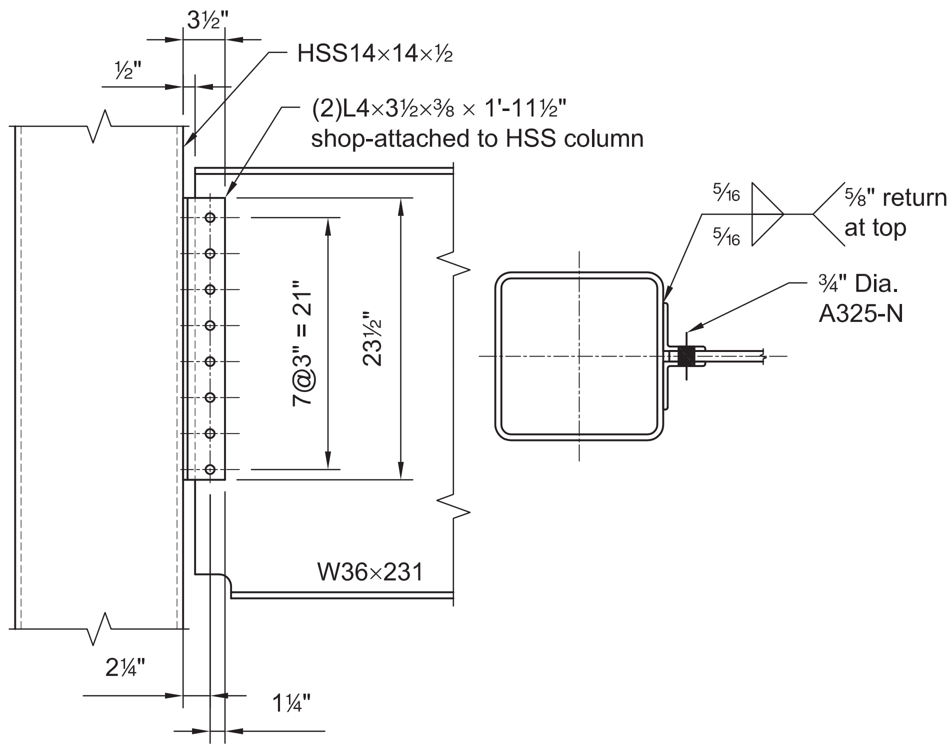
LRFD	ASD
$\phi = 0.75$ ϕR_n = 0.75(171 kips) = 128 kips 128 kips > 37.0 kips **o.k.**	$\Omega = 2.00$ $\dfrac{R_n}{\Omega} = \dfrac{171 \text{ kips}}{2.00}$ = 85.5 kips 85.5 kips > 24.7 kips **o.k.**

EXAMPLE K.3 DOUBLE ANGLE CONNECTION TO AN HSS COLUMN

Given:

Use AISC *Manual* Tables 10-1 and 10-2 to design a double-angle connection for an ASTM A992 W36×231 beam to an ASTM A500 Grade B HSS14×14×½ column. Use ¾-in.-diameter ASTM A325-N bolts in standard holes. The angles are ASTM A36 material. Use 70-ksi electrodes. The bottom flange cope is required for erection. Use the following vertical shear loads:

P_D = 37.5 kips
P_L = 113 kips



Solution:

From AISC *Manual* Table 2-4, the material properties are as follows:

Beam
ASTM A992
F_y = 50 ksi
F_u = 65 ksi

Column
ASTM A500 Grade B
F_y = 46 ksi
F_u = 58 ksi

Angles
ASTM A36
F_y = 36 ksi
F_u = 58 ksi

From AISC *Manual* Tables 1-1 and 1-12, the geometric properties are as follows:

W36×231
t_w = 0.760 in.
T = 31⅜ in.

HSS14×14×½
t = 0.465 in.
B = 14.0 in.

From Chapter 2 of ASCE/SEI 7, the required strength is:

LRFD	ASD
$R_u = 1.2(37.5 \text{ kips}) + 1.6(113 \text{ kips})$ $= 226 \text{ kips}$	$R_a = 37.5 \text{ kips} + 113 \text{ kips}$ $= 151 \text{ kips}$

Bolt and Weld Design

Try 8 rows of bolts and 5⁄16-in. welds.

Obtain the bolt group and angle available strength from AISC *Manual* Table 10-1.

LRFD	ASD
$\phi R_n = 286 \text{ kips} > 226 \text{ kips}$ **o.k.**	$\dfrac{R_n}{\Omega} = 191 \text{ kips} > 151 \text{ kips}$ **o.k.**

Obtain the available weld strength from AISC *Manual* Table 10-2 (welds B).

LRFD	ASD
$\phi R_n = 279 \text{ kips} > 226 \text{ kips}$ **o.k.**	$\dfrac{R_n}{\Omega} = 186 \text{ kips} > 151 \text{ kips}$ **o.k.**

Minimum Support Thickness

The minimum required support thickness using Table 10-2 is determined as follow for F_u = 58 ksi material.

$$0.238 \text{ in.}\left(\frac{65 \text{ ksi}}{58 \text{ ksi}}\right) = 0.267 \text{ in.} < 0.465 \text{ in.} \qquad \textbf{o.k.}$$

Minimum Angle Thickness

$t_{min} = w + {}^1\!/_{16}$ in., from AISC *Specification* Section J2.2b
$= {}^5\!/_{16} \text{ in.} + {}^1\!/_{16} \text{ in.}$
$= {}^3\!/_8 \text{ in.}$

Use ⅜-in. angle thickness to accommodate the welded legs of the double angle connection.

Use 2L4×32×⅜×1′-11½″.

Minimum Angle Length

$L = 23.5 \text{ in.} > T/2$

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> 31⅜ in./2
> 15.7 in. **o.k.**

Minimum Column Width

The workable flat for the HSS column is 14.0 in. – 2(2.25)(½ in.) = 11.8 in.

The recommended minimum shelf dimension for 5⁄16-in. fillet welds from AISC *Manual* Figure 8-11 is 9⁄16 in.

The minimum acceptable width to accommodate the connection is:

2(4.00 in.) + 0.760 in. + 2(9⁄16 in.) = 9.89 in. < 11.8 in. **o.k.**

Available Beam Web Strength

The available beam web strength, from AISC *Manual* Table 10-1 for an uncoped beam with L_{eh} = 1¾ in., is:

LRFD	ASD
ϕR_n = 702 kips/in.(0.760 in.) = 534 kips 534 kips > 226 kips **o.k.**	$\dfrac{R_n}{\Omega}$ = 468 kips/in.(0.760 in.) = 356 kips 356 kips > 151 kips **o.k.**

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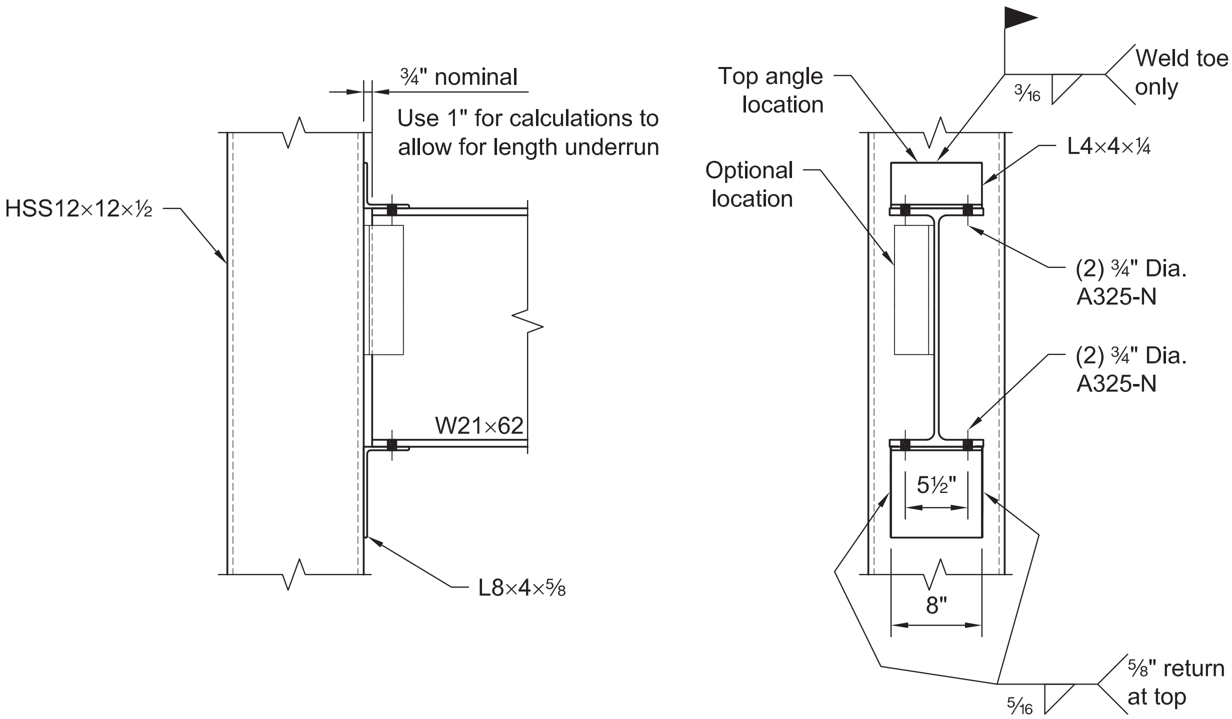
EXAMPLE K.4 UNSTIFFENED SEATED CONNECTION TO AN HSS COLUMN

Given:

Use AISC *Manual* Table 10-6 to design an unstiffened seated connection for an ASTM A992 W21×62 beam to an ASTM A500 Grade B HSS12×12×½ column. The angles are ASTM A36 material. Use 70-ksi electrodes. Use the following vertical shear loads:

P_D = 9.00 kips
P_L = 27.0 kips



Solution:

From AISC *Manual* Table 2-4, the material properties are as follows:

Beam
ASTM A992
F_y = 50 ksi
F_u = 65 ksi

Column
ASTM A500 Grade B
F_y = 46 ksi
F_u = 58 ksi

Angles
ASTM A36
F_y = 36 ksi
F_u = 58 ksi

From AISC *Manual* Tables 1-1 and 1-12, the geometric properties are as follows:

W21×62
t_w = 0.400 in.
d = 21.0 in.
k_{des} = 1.12 in.

HSS12×12×½
t = 0.465 in.
B = 12.0 in.

From Chapter 2 of ASCE/SEI 7, the required strength is:

LRFD	ASD
$R_u = 1.2(9.00\text{ kips}) + 1.6(27.0\text{ kips})$ $= 54.0\text{ kips}$	$R_a = 9.00\text{ kips} + 27.0\text{ kips}$ $= 36.0\text{ kips}$

Seat Angle and Weld Design

Check local web yielding of the W21×62 using AISC *Manual* Table 9-4 and Part 10.

LRFD	ASD
From AISC *Manual* Equation 9-45a and Table 9-4, $l_{b\,min} = \dfrac{R_u - \phi R_1}{\phi R_2} \geq k_{des}$ $= \dfrac{54.0\text{ kips} - 56.0\text{ kips}}{20.0\text{ kips/in.}} \geq 1.12\text{ in.}$ Use $l_{b\,min}$ = 1.12 in. Check web crippling when $l_b/d \leq 0.2$. From AISC *Manual* Equation 9-47a, $l_{b\,min} = \dfrac{R_u - \phi R_3}{\phi R_4}$ $= \dfrac{54.0\text{ kips} - 71.7\text{ kips}}{5.37\text{ kips/in.}}$ which results in a negative quantity. Check web crippling when $l_b/d > 0.2$. From AISC *Manual* Equation 9-48a, $l_{b\,min} = \dfrac{R_u - \phi R_5}{\phi R_6}$ $= \dfrac{54.0\text{ kips} - 64.2\text{ kips}}{7.16\text{ kips/in.}}$ which results in a negative quantity.	From AISC *Manual* Equation 9-45b and Table 9-4, $l_{b\,min} = \dfrac{R_a - R_1/\Omega}{R_2/\Omega} \geq k_{des}$ $= \dfrac{36.0\text{ kips} - 37.3\text{ kips}}{13.3\text{ kips/in.}} \geq 1.12\text{ in.}$ Use $l_{b\,min}$ = 1.12 in. Check web crippling when $l_b/d \leq 0.2$. From AISC *Manual* Equation 9-47b, $l_{b\,min} = \dfrac{R_a - R_3/\Omega}{R_4/\Omega}$ $= \dfrac{36.0\text{ kips} - 47.8\text{ kips}}{3.58\text{ kips/in.}}$ which results in a negative quantity. Check web crippling when $l_b/d > 0.2$. From AISC *Manual* Equation 9-48b, $l_{b\,min} = \dfrac{R_a - R_5/\Omega}{R_6/\Omega}$ $= \dfrac{36.0\text{ kips} - 42.8\text{ kips}}{4.77\text{ kips/in.}}$ which results in a negative quantity.

Note: Generally, the value of l_b/d is not initially known and the larger value determined from the web crippling equations in the preceding text can be used conservatively to determine the bearing length required for web crippling.

For this beam and end reaction, the beam web strength exceeds the required strength (hence the negative bearing lengths) and the lower-bound bearing length controls ($l_{b\,req} = k_{des}$ = 1.12 in.). Thus, $l_{b\,min}$ = 1.12 in.

Try an L8×4×⅝ seat with 5⁄16-in. fillet welds.

Outstanding Angle Leg Available Strength

From AISC *Manual* Table 10-6 for an 8-in. angle length and $l_{b,req}$ = 1.12 in.≈1⅛ in., the outstanding angle leg available strength is:

LRFD		ASD	
$\phi R_n = 81.0\text{ kips} > 54.0\text{ kips}$	**o.k.**	$\dfrac{R_n}{\Omega} = 53.9\text{ kips} > 36.0\text{ kips}$	**o.k.**

Available Weld Strength

From AISC *Manual* Table 10-6, for an 8 in. x 4 in. angle and 5⁄16-in. weld size, the available weld strength is:

LRFD		ASD	
$\phi R_n = 66.7\text{ kips} > 54.0\text{ kips}$	**o.k.**	$\dfrac{R_n}{\Omega} = 44.5\text{ kips} > 36.0\text{ kips}$	**o.k.**

Minimum HSS Wall Thickness to Match Weld Strength

$$t_{min} = \frac{3.09D}{F_u} \qquad \text{(}\textit{Manual}\text{ Eq. 9-2)}$$

$$= \frac{3.09(5)}{58\text{ ksi}}$$

$$= 0.266\text{ in.} < 0.465\text{ in.}$$

Because t of the HSS is greater than t_{min} for the 5⁄16-in. weld, no reduction in the weld strength is required to account for the shear in the HSS.

Connection to Beam and Top Angle (AISC Manual Part 10)

Use a L4×4×¼ top angle. Use a 3⁄16-in. fillet weld across the toe of the angle for attachment to the HSS. Attach both the seat and top angles to the beam flanges with two ¾-in.-diameter ASTM A325-N bolts.

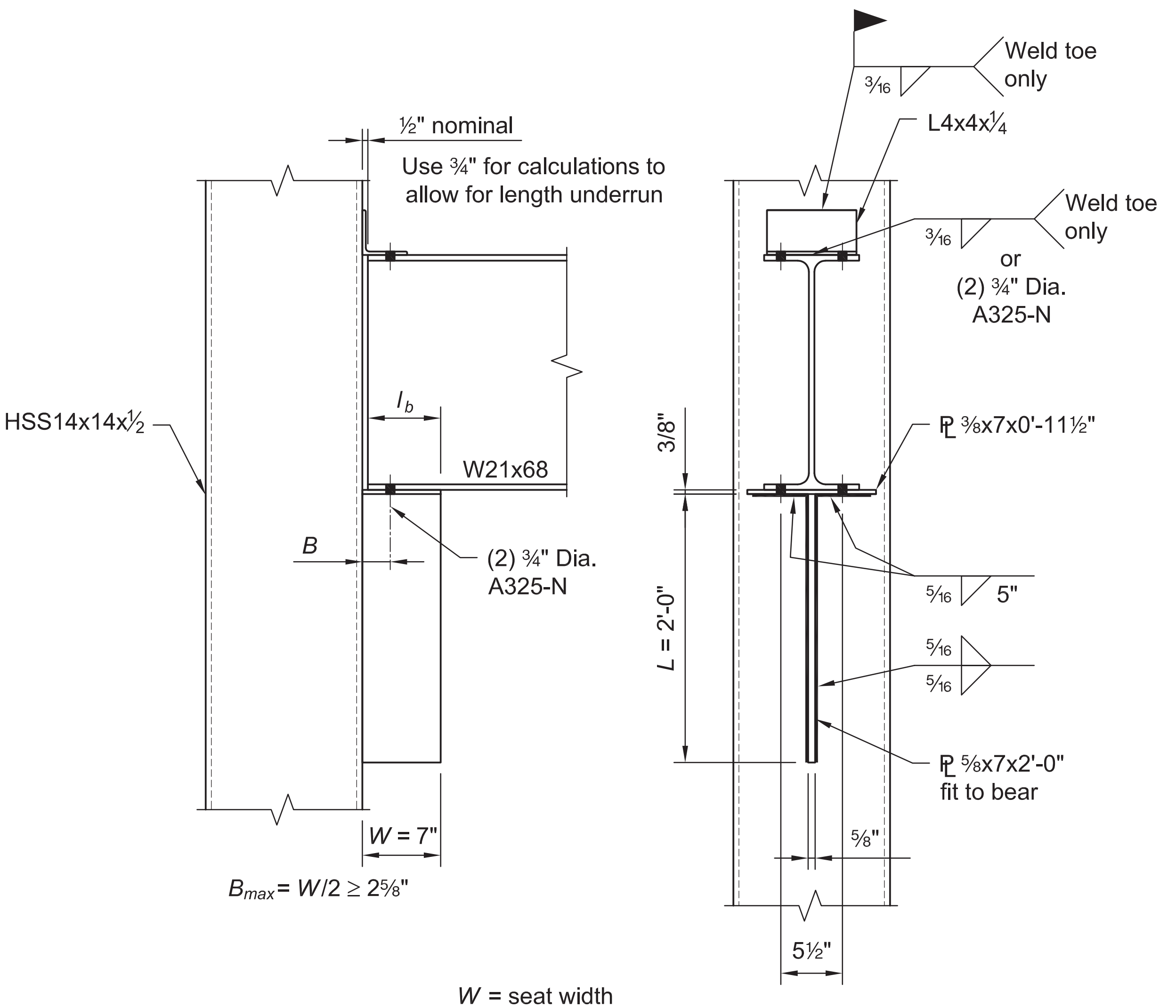
EXAMPLE K.5 STIFFENED SEATED CONNECTION TO AN HSS COLUMN

Given:

Use AISC *Manual* Tables 10-8 and 10-15 to design a stiffened seated connection for an ASTM A992 W21×68 beam to an ASTM A500 Grade B HSS14×14×½ column. Use the following vertical shear loads:

P_D = 20.0 kips
P_L = 60.0 kips

Use ¾-in.-diameter ASTM A325-N bolts in standard holes to connect the beam to the seat plate. Use 70-ksi electrode welds to connect the stiffener, seat plate and top angle to the HSS. The angle and plate material is ASTM A36.



Solution:

From AISC *Manual* Tables 2-4 and 2-5, the material properties are as follows:

Beam
ASTM A992
$F_y = 50$ ksi
$F_u = 65$ ksi

Column
ASTM A500 Grade B
$F_y = 46$ ksi
$F_u = 58$ ksi

Angles and Plates
ASTM A36
$F_y = 36$ ksi
$F_u = 58$ ksi

From AISC *Manual* Tables 1-1 and 1-12, the geometric properties are as follows:

W21×68
$t_w = 0.430$ in.
$d = 21.1$ in.
$k_{des} = 1.19$ in.

HSS14×14×½
$t = 0.465$ in.
$B = 14.0$ in.

From Chapter 2 of ASCE/SEI 7, the required strength is:

LRFD	ASD
$P_u = 1.2(20.0 \text{ kips}) + 1.6(60.0 \text{ kips})$ $= 120 \text{ kips}$	$P_a = 20.0 \text{ kips} + 60.0 \text{ kips}$ $= 80.0 \text{ kips}$

Limits of Applicability for AISC Specification Section K1.3

AISC *Specification* Table K1.2A gives the limits of applicability for plate-to-rectangular HSS connections. The limits that are applicable here are:

Strength: $F_y = 46 \text{ ksi} \leq 52 \text{ ksi}$ **o.k.**

Ductility: $\dfrac{F_y}{F_u} = \dfrac{46 \text{ ksi}}{58 \text{ ksi}}$

$= 0.793 \leq 0.8$ **o.k.**

Stiffener Width, W, Required for Web Local Crippling and Web Local Yielding

The stiffener width is determined based on web local crippling and web local yielding of the beam.

For web local crippling, assume $l_b/d > 0.2$ and use constants R_5 and R_6 from AISC *Manual* Table 9-4. Assume a ¾-in. setback for the beam end.

LRFD	ASD
$\phi R_5 = 75.9\text{ kips}$ $\phi R_6 = 7.95\text{ kips}$ $W_{min} = \dfrac{R_u - \phi R_5}{\phi R_6} + \text{setback} \geq k_{des} + \text{setback}$ $= \dfrac{120\text{ kips} - 75.9\text{ kips}}{7.95\text{ kips/in.}} + \text{¾ in.} \geq 1.19\text{ in.} + \text{¾ in.}$ $= 6.30\text{ in.} \geq 1.94\text{ in.}$	$R_5/\Omega = 50.6\text{ kips}$ $R_6/\Omega = 5.30\text{ kips}$ $W_{min} = \dfrac{R_a - R_5/\Omega}{R_6/\Omega} + \text{setback} \geq k_{des} + \text{setback}$ $= \dfrac{80.0\text{ kips} - 50.6\text{ kips}}{5.30\text{ kips/in.}} + \text{¾ in.} \geq 1.19\text{ in.} + \text{¾ in.}$ $= 6.30\text{ in.} \geq 1.94\text{ in.}$

For web local yielding, use constants R_1 and R_2 from AISC *Manual* Table 9-4. Assume a ¾-in. setback for the beam end.

LRFD	ASD
$\phi R_1 = 64.0\text{ kips}$ $\phi R_2 = 21.5\text{ kips}$ $W_{min} = \dfrac{R_u - \phi R_1}{\phi R_2} + \text{setback}$ $= \dfrac{120\text{ kips} - 64.0\text{ kips}}{21.5\text{ kips/in.}} + \text{¾ in.}$ $= 3.35\text{ in.}$	$R_1/\Omega = 42.6\text{ kips}$ $R_2/\Omega = 14.3\text{ kips}$ $W_{min} = \dfrac{R_a - R_1/\Omega}{R_2/\Omega} + \text{setback}$ $= \dfrac{80.0\text{ kips} - 42.6\text{ kips}}{14.3\text{ kips/in.}} + \text{¾ in.}$ $= 3.37\text{ in.}$

The minimum stiffener width, W_{min}, for web local crippling controls. Use W = 7.00 in.

Check the assumption that $l_b/d > 0.2$.

$$
\begin{aligned}
l_b &= 7.00\text{ in.} - \text{¾ in.} \\
&= 6.25\text{ in.} \\
\frac{l_b}{d} &= \frac{6.25\text{ in.}}{21.1\text{ in.}} \\
&= 0.296 > 0.2\text{, as assumed}
\end{aligned}
$$

Weld Strength Requirements for the Seat Plate

Try a stiffener of length, L = 24 in. with 5⁄16-in. fillet welds. Enter AISC *Manual* Table 10-8 using W = 7 in. as determined in the preceding text.

LRFD	ASD
$\phi R_n = 293\text{ kips} > 120\text{ kips}$ **o.k.**	$\dfrac{R_n}{\Omega} = 195\text{ kips} > 80.0\text{ kips}$ **o.k.**

From AISC *Manual* Part 10, Figure 10-10(b), the minimum length of the seat-plate-to-HSS weld on each side of the stiffener is $0.2L$ = 4.8 in. This establishes the minimum weld between the seat plate and stiffener; use 5 in. of 5⁄16-in. weld on each side of the stiffener.

Minimum HSS Wall Thickness to Match Weld Strength

The minimum HSS wall thickness required to match the shear rupture strength of the base metal to that of the weld is:

$$t_{min} = \frac{3.09D}{F_u} \qquad \text{(\textit{Manual} Eq. 9-2)}$$
$$= \frac{3.09(5)}{58 \text{ ksi}}$$
$$= 0.266 \text{ in.} < 0.465 \text{ in.}$$

Because t of the HSS is greater than t_{min} for the 5⁄16-in. fillet weld, no reduction in the weld strength to account for shear in the HSS is required.

Stiffener Plate Thickness

From AISC *Manual* Part 10, Design Table 10-8 discussion, to develop the stiffener-to-seat-plate welds, the minimum stiffener thickness is:

$$t_{p\,min} = 2w$$
$$= 2(5/16 \text{ in.})$$
$$= 5/8 \text{ in.}$$

For a stiffener with F_y = 36 ksi and a beam with F_y = 50 ksi, the minimum stiffener thickness is:

$$t_{p\,min} = \left(\frac{F_{y\,beam}}{F_{y\,stiffener}}\right)t_w$$
$$= \left(\frac{50 \text{ ksi}}{36 \text{ ksi}}\right)(0.430 \text{ in.})$$
$$= 0.597 \text{ in.}$$

For a stiffener with F_y = 36 ksi and a column with F_u = 58 ksi, the maximum stiffener thickness is determined from AISC *Specification* Table K1.2 as follows:

$$t_{p\,max} = \frac{F_u t}{F_{yp}} \qquad \text{(from \textit{Spec}. Eq. K1-3)}$$
$$= \frac{58 \text{ ksi}(0.465 \text{ in.})}{36 \text{ ksi}}$$
$$= 0.749 \text{ in.}$$

Use stiffener thickness of 5⁄8 in.

Determine the stiffener length using AISC *Manual* Table 10-15.

LRFD	ASD
$\left(\frac{R_u W}{t^2}\right)_{req} = \frac{120 \text{ kips}(7.00 \text{ in.})}{(0.465 \text{ in.})^2}$ $= 3{,}880 \text{ kips/in.}$	$\left(\frac{R_a W}{t^2}\right)_{req} = \frac{80.0 \text{ kips}(7.00 \text{ in.})}{(0.465 \text{ in.})^2}$ $= 2{,}590 \text{ kips/in.}$

To satisfy the minimum, select a stiffener L = 24 in. from AISC *Manual* Table 10-15.

LRFD	ASD
$\dfrac{R_u W}{t^2}$ = 3,910 kips/in. > 3,880 kips/in. **o.k.**	$\dfrac{R_a W}{t^2}$ = 2,600 kips/in. > 2,590 kips/in. **o.k.**

Use PL⅝ in.×7 in.×2 ft-0 in. for the stiffener.

HSS Width Check

The minimum width is 0.4L + t_p + 2(2.25t)

$$\begin{aligned} B = 14.0 \text{ in.} &> 0.4(24.0 \text{ in.}) + \tfrac{5}{8} \text{ in.} + 2(2.25)(0.465 \text{ in.}) \\ &= 12.3 \text{ in.} \end{aligned}$$

Seat Plate Dimensions

To accommodate two ¾-in.-diameter ASTM A325-N bolts on a 5½-in. gage connecting the beam flange to the seat plate, a width of 8 in. is required. To accommodate the seat-plate-to-HSS weld, the required width is:

$$2(5.00 \text{ in.}) + \tfrac{5}{8} \text{ in.} = 10.6 \text{ in.}$$

Note: To allow room to start and stop welds, an 11.5 in. width is used.

Use PL⅜ in.×7 in.×0 ft-11½ in. for the seat plate.

Top Angle, Bolts and Welds (AISC Manual Part 10)

The minimum weld size for the HSS thickness according to AISC *Specification* Table J2.4 is ³⁄₁₆ in. The angle thickness should be ¹⁄₁₆ in. larger.

Use L4×4×¼ with ³⁄₁₆-in. fillet welds along the toes of the angle to the beam flange and HSS. Alternatively, two ¾-in.-diameter ASTM A325-N bolts may be used to connect the leg of the angle to the beam flange.

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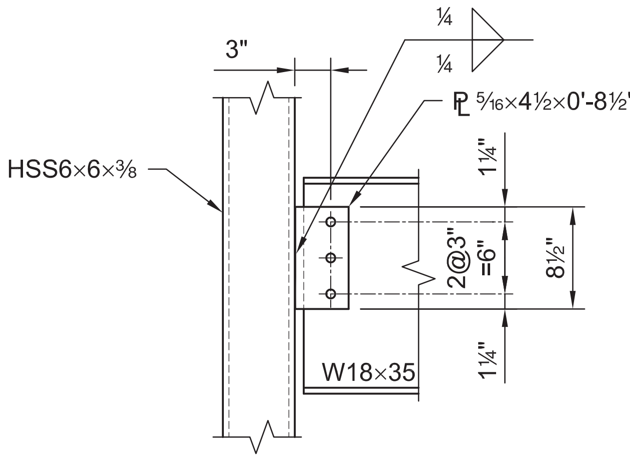
EXAMPLE K.6 SINGLE-PLATE CONNECTION TO A RECTANGULAR HSS COLUMN

Given:

Use AISC *Manual* Table 10-10a to design a single-plate connection for an ASTM A992 W18×35 beam framing into an ASTM A500 Grade B HSS6×6×⅜ column. Use ¾-in.-diameter ASTM A325-N bolts in standard holes and 70-ksi weld electrodes. The plate material is ASTM A36. Use the following vertical shear loads:

P_D = 6.50 kips
P_L = 19.5 kips



Solution:

From AISC *Manual* Tables 2-4 and 2-5, the material properties are as follows:

Beam
ASTM A992
F_y = 50 ksi
F_u = 65 ksi

Column
ASTM A500 Grade B
F_y = 46 ksi
F_u = 58 ksi

Plate
ASTM A36
F_y = 36 ksi
F_u = 58 ksi

From AISC *Manual* Tables 1-1 and 1-12, the geometric properties are as follows:

W18×35
d = 17.7 in.
t_w = 0.300 in.
T = 15½ in.

HSS6×6×⅜
$B = H = 6.00$ in.
$t = 0.349$ in.
$b/t = 14.2$

From Chapter 2 of ASCE/SEI 7, the required strength is:

LRFD	ASD
$R_u = 1.2(6.50\text{ kips}) + 1.6(19.5\text{ kips})$ $= 39.0\text{ kips}$	$R_a = 6.50\text{ kips} + 19.5\text{ kips}$ $= 26.0\text{ kips}$

Limits of Applicability of AISC Specification Section K1.3

AISC *Specification* Table K1.2A gives the following limits of applicability for plate-to-rectangular HSS connections. The limits applicable here are:

HSS wall slenderness:

$$\frac{(B-3t)}{t} \le 1.40\sqrt{\frac{E}{F_y}}$$

$$\frac{[6.00\text{ in.} - 3(0.349\text{ in.})]}{0.349\text{ in.}} \le 1.40\sqrt{\frac{29{,}000\text{ ksi}}{46\text{ ksi}}}$$

$14.2 < 35.2$ **o.k.**

Material Strength:
$F_y = 46\text{ ksi} \le 52\text{ ksi}$ **o.k.**

Ductility:

$$\frac{F_y}{F_u} = \frac{46\text{ ksi}}{58\text{ ksi}}$$
$$= 0.793 \le 0.8$$ **o.k.**

Using AISC *Manual* Part 10, determine if a single-plate connection is suitable (the HSS wall is not slender).

Maximum Single-Plate Thickness

From AISC *Specification* Table K1.2, the maximum single-plate thickness is:

$$t_p \le \frac{F_u}{F_{yp}}t \qquad (Spec.\text{ Eq. K1-3})$$
$$= \frac{58\text{ ksi}}{36\text{ ksi}}(0.349\text{ in.})$$
$$= 0.562\text{ in.}$$

Note: Limiting the single-plate thickness precludes a shear yielding failure of the HSS wall.

Single-Plate Connection

Try 3 bolts and a 5⁄16-in. plate thickness with ¼-in. fillet welds.

$t_p = 5/16\text{ in.} < 0.562\text{ in.}$ **o.k.**

Note: From AISC *Manual* Table 10-9, either the plate or the beam web must satisfy:

$$
\begin{aligned}
t = \tfrac{5}{16}\text{ in.} &\le d/2 + \tfrac{1}{16}\text{ in.} \\
&\le \tfrac{3}{4}\text{ in.}/2 + \tfrac{1}{16}\text{ in.} \\
&\le 0.438\text{ in.} \qquad \textbf{o.k.}
\end{aligned}
$$

Obtain the available single-plate connection strength from AISC *Manual* Table 10-10a.

LRFD	ASD
$\phi R_n = 43.4\text{ kips} > 39.0\text{ kips}$ **o.k.**	$\dfrac{R_n}{\Omega} = 28.8\text{ kips} > 26.0\text{ kips}$ **o.k.**

Use a PL5⁄16×4½×0′-8½″.

HSS Shear Rupture at Welds

The minimum HSS wall thickness required to match the shear rupture strength of the HSS wall to that of the weld is:

$$
\begin{aligned}
t_{min} &= \frac{3.09D}{F_u} && \text{(}\textit{Manual}\text{ Eq. 9-2)} \\
&= \frac{3.09(4)}{58\text{ ksi}} \\
&= 0.213\text{ in.} < t = 0.349\text{ in.} \qquad \textbf{o.k.}
\end{aligned}
$$

Available Beam Web Bearing Strength (AISC Manual Table 10-1)

For three ¾-in.-diameter bolts and L_{eh} = 1½ in., the bottom of AISC *Manual* Table 10-1 gives the uncoped beam web available bearing strength per inch of thickness. The available beam web bearing strength can then be calculated as follows:

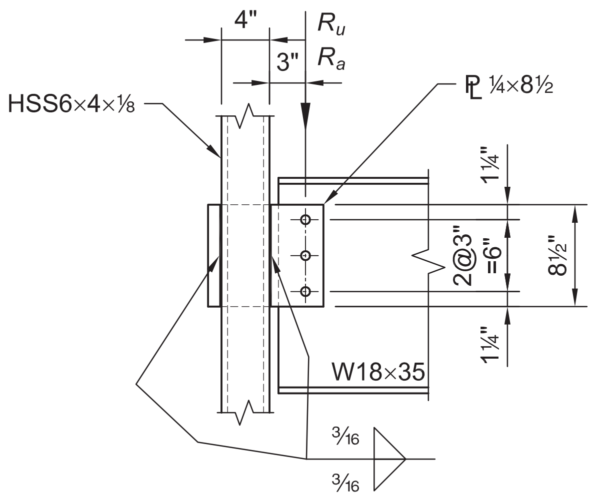
LRFD	ASD
$\phi R_n = 263\text{ kips/in.}(0.300\text{ in.})$ $= 78.9\text{ kips} > 39.0\text{ kips}$ **o.k.**	$\dfrac{R_n}{\Omega} = 176\text{ kips/in.}(0.300\text{ in.})$ $= 52.8\text{ kips} > 26.0\text{ kips}$ **o.k.**

EXAMPLE K.7 THROUGH-PLATE CONNECTION

Given:

Use AISC *Manual* Table 10-10a to check a through-plate connection between an ASTM A992 W18×35 beam and an ASTM A500 Grade B HSS6×4×⅛ with the connection to one of the 6 in. faces, as shown in the figure. A thin-walled column is used to illustrate the design of a through-plate connection. Use ¾-in.-diameter ASTM A325-N bolts in standard holes and 70-ksi weld electrodes. The plate is ASTM A36 material. Use the following vertical shear loads:

P_D = 3.30 kips
P_L = 9.90 kips



Solution:

From AISC *Manual* Tables 2-4 and 2-5, the material properties are as follows:

Beam
ASTM A992
F_y = 50 ksi
F_u = 65 ksi

Column
ASTM A500 Grade B
F_y = 46 ksi
F_u = 58 ksi

Plate
ASTM A36
F_y = 36 ksi
F_u = 58 ksi

From AISC *Manual* Tables 1-1 and 1-11, the geometric properties are as follows:

W18×35
d = 17.7 in.
t_w = 0.300 in.
T = 15½ in.

HSS6×4×⅛
B = 4.00 in.
H = 6.00 in.
t = 0.116 in.
h/t = 48.7

Limits of Applicability of AISC Specification Section K1.3

AISC *Specification* Table K1.2A gives the following limits of applicability for plate-to-rectangular HSS connections. The limits applicable here follow.

HSS wall slenderness: Check if a single-plate connection is allowed.

$$\frac{(H-3t)}{t} \leq 1.40\sqrt{\frac{E}{F_y}}$$

$$\frac{\left[6.00-3(0.116\,\text{in.})\right]}{0.116\,\text{in.}} \leq 1.40\sqrt{\frac{29{,}000\text{ ksi}}{46\text{ ksi}}}$$

$48.7 > 35.2$ **n.g.**

Because the HSS6×4×⅛ is slender, a through-plate connection should be used instead of a single-plate connection. Through-plate connections are typically very expensive. When a single-plate connection is not adequate, another type of connection, such as a double-angle connection may be preferable to a through-plate connection.

AISC *Specification* Chapter K does not contain provisions for the design of through-plate shear connections. The following procedure treats the connection of the through-plate to the beam as a single-plate connection.

From Chapter 2 of ASCE/SEI 7, the required strength is:

LRFD	ASD
$R_u = 1.2(3.30\text{ kips}) + 1.6(9.90\text{ kips})$ $= 19.8\text{ kips}$	$R_a = 3.30\text{ kips} + 9.90\text{ kips}$ $= 13.2\text{ kips}$

Portion of the Through-Plate Connection that Resembles a Single-Plate

Try three rows of bolts (L = 8½ in.) and a ¼-in. plate thickness with ³⁄₁₆-in. fillet welds.

$$\begin{aligned} L = 8\tfrac{1}{2}\text{ in.} &\geq T/2 \\ &\geq (15\tfrac{1}{2}\text{ in.})/2 \\ &\geq 7.75\text{ in.} \end{aligned}$$ **o.k.**

Note: From AISC *Manual* Table 10-9, either the plate or the beam web must satisfy:

$$\begin{aligned} t = \tfrac{1}{4}\text{ in.} &\leq d_b/2 + \tfrac{1}{16}\text{ in.} \\ &\leq \tfrac{3}{4}\text{ in.}/2 + \tfrac{1}{16}\text{ in.} \\ &\leq 0.438\text{ in.} \end{aligned}$$ **o.k.**

Obtain the available single-plate connection strength from AISC *Manual* Table 10-10a.

LRFD	ASD
$\phi R_n = 38.3\text{ kips} > 19.8\text{ kips}$ **o.k.**	$\dfrac{R_n}{\Omega} = 25.6\text{ kips} > 13.2\text{ kips}$ **o.k.**

Required Weld Strength

The available strength for the welds in this connection is checked at the location of the maximum reaction, which is along the weld line closest to the bolt line. The reaction at this weld line is determined by taking a moment about the weld line farthest from the bolt line.

a = 3.00 in. (distance from bolt line to nearest weld line)

LRFD	ASD
$V_{fu} = \dfrac{R_u\left(B+a\right)}{B}$	$V_{fa} = \dfrac{R_a\left(B+a\right)}{B}$
$= \dfrac{19.8\text{ kips}\left(4.00\text{ in.}+3.00\text{ in.}\right)}{4.00\text{ in.}}$	$= \dfrac{13.2\text{ kips}\left(4.00\text{ in.}+3.00\text{ in.}\right)}{4.00\text{ in.}}$
$= 34.7\text{ kips}$	$= 23.1\text{ kips}$

Available Weld Strength

The minimum required weld size is determined using AISC *Manual* Part 8.

LRFD	ASD
$D_{req} = \dfrac{\phi R_n}{1.392l}$ (from *Manual* Eq. 8-2a)	$D_{req} = \dfrac{R_n/\Omega}{0.928l}$ (from *Manual* Eq. 8-2b)
$= \dfrac{34.7\text{ kips}}{1.392\left(8.50\text{ in.}\right)\left(2\right)}$	$= \dfrac{23.1\text{ kips}}{0.928\left(8.50\text{ in.}\right)\left(2\right)}$
$= 1.47\text{ sixteenths} < 3\text{ sixteenths}$ **o.k.**	$= 1.46\text{ sixteenths} < 3\text{ sixteenths}$ **o.k.**

HSS Shear Yielding and Rupture Strength

The available shear strength of the HSS due to shear yielding and shear rupture is determined from AISC *Specification* Section J4.2.

LRFD	ASD
For shear yielding of HSS at the connection,	For shear yielding of HSS at the connection,
$\phi = 1.00$	$\Omega = 1.50$
$\phi R_n = \phi 0.60 F_y A_{gv}$ (from *Spec.* Eq. J4-3)	$\dfrac{R_n}{\Omega} = \dfrac{0.60 F_y A_{gv}}{\Omega}$ (from *Spec.* Eq. J4-3)
$= 1.00\left(0.60\right)\left(46\text{ ksi}\right)\left(0.116\text{ in.}\right)\left(8.50\text{ in.}\right)\left(2\right)$	$= \dfrac{\left(0.60\right)\left(46\text{ ksi}\right)\left(0.116\text{ in.}\right)\left(8.50\text{ in.}\right)\left(2\right)}{1.50}$
$= 54.4\text{ kips}$	$= 36.3\text{ kips}$
54.4 kips > 34.7 kips **o.k.**	36.3 kips > 23.1 kips **o.k.**
For shear rupture of HSS at the connection,	For shear rupture of HSS at the connection,
$\phi = 0.75$	$\Omega = 2.00$

$\phi R_n = \phi 0.60 F_u A_{nv}$ (from *Spec.* Eq. J4-4) $= 0.75(0.60)(58\text{ ksi})(0.116\text{ in.})(8.50\text{ in.})(2)$ $= 51.5\text{ kips}$ 51.5 kips > 34.7 kips **o.k.**	$\dfrac{R_n}{\Omega} = \dfrac{0.60 F_u A_{nv}}{\Omega}$ (from *Spec.* Eq. J4-4) $= \dfrac{(0.60)(58\text{ ksi})(0.116\text{ in.})(8.50\text{ in.})(2)}{2.00}$ $= 34.3\text{ kips}$ 34.3 kips > 23.1 kips **o.k.**

Available Beam Web Bearing Strength

The available beam web bearing strength is determined from the bottom portion of Table 10-1, for three ¾-in.-diameter bolts and an uncoped beam. The table provides the available strength in kips/in. and the available beam web bearing strength is:

LRFD	ASD
$\phi R_n = 263\text{ kips/in.}(0.300\text{ in.})$ $= 78.9\text{ kips} > 19.8\text{ kips}$ **o.k.**	$\dfrac{R_n}{\Omega} = 176\text{ kips/in.}(0.300\text{ in.})$ $= 52.8\text{ kips} > 13.2\text{ kips}$ **o.k.**

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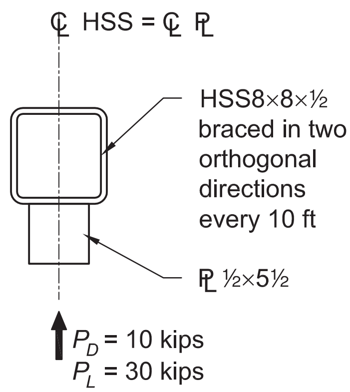
EXAMPLE K.8 TRANSVERSE PLATE LOADED PERPENDICULAR TO THE HSS AXIS ON A RECTANGULAR HSS

Given:

Verify the local strength of the ASTM A500 Grade B HSS column subject to the transverse loads given as follows, applied through a 5½-in.-wide ASTM A36 plate. The HSS8×8×½ is in compression with nominal axial loads of $P_{D\ column}$ = 54.0 kips and $P_{L\ column}$ = 162 kips. The HSS has negligible required flexural strength.



Solution:

From AISC *Manual* Tables 2-4 and 2-5, the material properties are as follows:

Column
ASTM A500 Grade B
F_y = 46 ksi
F_u = 58 ksi

Plate
ASTM A36
F_{yp} = 36 ksi
F_u = 58 ksi

From AISC *Manual* Table 1-12, the geometric properties are as follows:

HSS8×8×½
B = 8.00 in.
t = 0.465 in.

Plate
B_p = 5½ in.
t_p = ½ in.

Limits of Applicability of AISC Specification Section K1.3

AISC *Specification* Table K1.2A provides the limits of applicability for plate-to-rectangular HSS connections. The following limits are applicable in this example.

HSS wall slenderness:
$B/t = 14.2 \leq 35$ **o.k.**

Width ratio:

$B_p/B = 5\frac{1}{2}$ in./8.00 in.
$= 0.688$

$0.25 \leq B_p/B \leq 1.0$ **o.k.**

Material strength:

$F_y = 46$ ksi ≤ 52 ksi for HSS **o.k.**

Ductility:

$F_y/F_u = 46$ ksi/58 ksi
$= 0.793 \leq 0.8$ for HSS **o.k.**

From Chapter 2 of ASCE/SEI 7, the required strength is:

LRFD	ASD
Transverse force from the plate: $P_u = 1.2(10.0 \text{ kips}) + 1.6(30.0 \text{ kips})$ $= 60.0$ kips Column axial force: $P_r = P_{u\ column}$ $= 1.2(54.0 \text{ kips}) + 1.6(162 \text{ kips})$ $= 324$ kips	Transverse force from the plate: $P_a = 10.0 \text{ kips} + 30.0 \text{ kips}$ $= 40.0$ kips Column axial force: $P_r = P_{a\ column}$ $= 54.0 \text{ kips} + 162 \text{ kips}$ $= 216$ kips

Available Local Yielding Strength of Plate from AISC Specification Table K1.2

The available local yielding strength of the plate is determined from AISC *Specification* Table K1.2.

$$R_n = \frac{10}{B/t} F_y t B_p \leq F_{yp} t_p B_p \qquad (\textit{Spec}.\ \text{Eq. K1-7})$$

$$= \frac{10}{8.00 \text{ in.}/0.465 \text{ in.}}(46 \text{ ksi})(0.465 \text{ in.})(5\tfrac{1}{2} \text{ in.}) \leq 36 \text{ ksi}(\tfrac{1}{2} \text{ in.})(5\tfrac{1}{2} \text{ in.})$$

$= 68.4$ kips ≤ 99.0 kips **o.k.**

LRFD	ASD
$\phi = 0.95$ $\phi R_n = 0.95(68.4 \text{ kips})$ $= 65.0 \text{ kips} > 60.0 \text{ kips}$ **o.k.**	$\Omega = 1.58$ $\dfrac{R_n}{\Omega} = \dfrac{68.4 \text{ kips}}{1.58}$ $= 43.3 \text{ kips} > 40.0 \text{ kips}$ **o.k.**

HSS Shear Yielding (Punching)

The available shear yielding (punching) strength of the HSS is determined from AISC *Specification* Table K1.2.

This limit state need not be checked when $B_p > B - 2t$, nor when $B_p < 0.85B$.

$B - 2t = 8.00$ in. $- 2(0.465$ in.$)$
$= 7.07$ in.

$0.85B = 0.85(8.00$ in.$)$
$= 6.80$ in.

Therefore, because B_p < 6.80 in., this limit state does not control.

Other Limit States

The other limit states listed in AISC *Specification* Table K1.2 apply only when β = 1.0. Because $B_p/B < 1.0$, these limit states do not apply.

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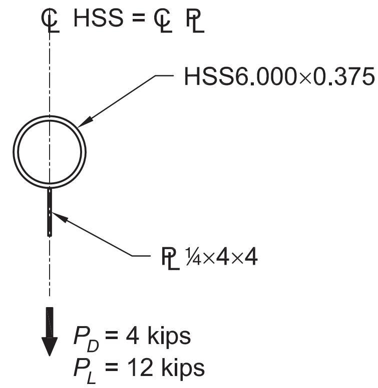
EXAMPLE K.9 LONGITUDINAL PLATE LOADED PERPENDICULAR TO THE HSS AXIS ON A ROUND HSS

Given:

Verify the local strength of the ASTM A500 Grade B HSS6.000×0.375 tension chord subject to transverse loads, P_D = 4.00 kips and P_L = 12.0 kips, applied through a 4 in. wide ASTM A36 plate.



Solution:

From AISC *Manual* Tables 2-4 and 2-5, the material properties are as follows:

Chord
ASTM A500 Grade B
F_y = 42 ksi
F_u = 58 ksi

Plate
ASTM A36
F_{yp} = 36 ksi
F_u = 58 ksi

From AISC *Manual* Table 1-13, the geometric properties are as follows:

HSS6.000×0.375
D = 6.00 in.
t = 0.349 in.

Limits of Applicability of AISC Specification Section K1.2, Table K1.1A

AISC *Specification* Table K1.1A provides the limits of applicability for plate-to-round connections. The applicable limits for this example are:

HSS wall slenderness:

D/t = 6.00 in./0.349 in.
= 17.2 ≤ 50 for T-connections **o.k.**

Material strength:

F_y = 42 ksi ≤ 52 ksi for HSS **o.k.**

Ductility:

F_y/F_u = 42 ksi/58 ksi
= 0.724 ≤ 0.8 for HSS **o.k.**

From Chapter 2 of ASCE/SEI 7, the required strength is:

LRFD	ASD
P_u = 1.2(4.00 kips) + 1.6(12.0 kips) = 24.0 kips	P_a = 4.00 kips + 12.0 kips = 16.0 kips

HSS Plastification Limit State

The limit state of HSS plastification applies and is determined from AISC *Specification* Table K1.1.

$$R_n = 5.5F_y t^2\left(1+0.25\frac{l_b}{D}\right)Q_f$$ (*Spec*. Eq. K1-2)

From the AISC *Specification* Table K1.1 Functions listed at the bottom of the table, for an HSS connecting surface in tension, Q_f = 1.0.

$$\begin{aligned}R_n &= 5.5(42\text{ ksi})(0.349\text{ in.})^2\left(1+0.25\frac{4.00\text{ in.}}{6.00\text{ in.}}\right)(1.0)\\ &= 32.8\text{ kips}\end{aligned}$$

The available strength is:

LRFD	ASD
ϕR_n = 0.90(32.8 kips) = 29.5 kips > 24.0 kips **o.k.**	$\dfrac{R_n}{\Omega} = \dfrac{32.8\text{ kips}}{1.67}$ = 19.6 kips > 16.0 kips **o.k.**

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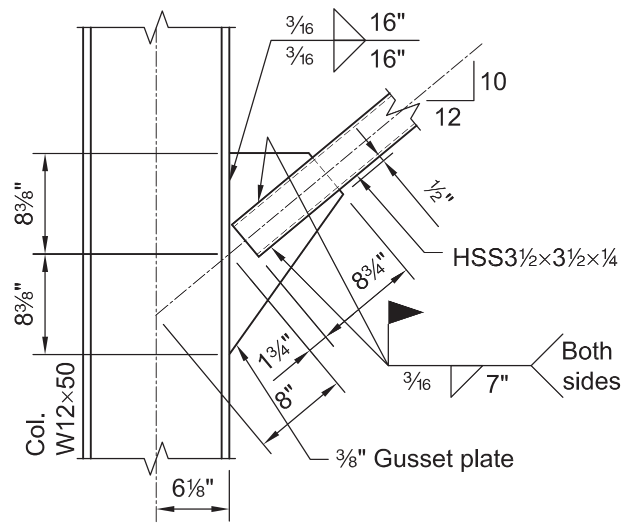
EXAMPLE K.10 HSS BRACE CONNECTION TO A W-SHAPE COLUMN

Given:

Verify the strength of an ASTM A500 Grade B HSS3½×3½×¼ brace for required axial forces of 80.0 kips (LRFD) and 52.0 kips (ASD). The axial force may be either tension or compression. The length of the brace is 6 ft. Design the connection of the HSS brace to the gusset plate. Allow 1⁄16 in. for fit of the slot over the gusset plate.



Solution:

From AISC *Manual* Tables 2-4 and 2-5, the material properties are as follows:

Brace
ASTM A500 Grade B
F_y = 46 ksi
F_u = 58 ksi

Plate
ASTM A36
F_{yp} = 36 ksi
F_u = 58 ksi

From AISC *Manual* Table 1-12, the geometric properties are as follows:

HSS3½×3½×¼
A = 2.91 in.^{2}
r = 1.32 in.
t = 0.233 in.

Available Compressive Strength of Brace

Obtain the available axial compressive strength of the brace from AISC *Manual* Table 4-4.

K = 1.0
L_b = 6.00 ft

LRFD		ASD	
$\phi_c P_n = 98.6$ kips > 80.0 kips	**o.k.**	$\dfrac{P_n}{\Omega_c} = 65.6$ kips > 52.0 kips	**o.k.**

Available Tensile Yielding Strength of Brace

Obtain the available tensile yielding strength of the brace from AISC *Manual* Table 5-5.

LRFD		ASD	
$\phi_t P_n = 120$ kips > 80.0 kips	**o.k.**	$\dfrac{P_n}{\Omega_t} = 80.2$ kips > 52.0 kips	**o.k.**

Available Tensile Rupture Strength of the Brace

Due to plate geometry, 8¾ in. of overlap occurs. Try four 3⁄16-in. fillet welds, each 7-in. long. Based on AISC *Specification* Table J2.4 and the HSS thickness of ¼ in., the minimum weld size is an ⅛-in. fillet weld.

Determine the available tensile strength of the brace from AISC *Specification* Section D2.

$$A_e = A_n U \qquad \text{(Spec. Eq. D3-1)}$$

where

$$\begin{aligned} A_n &= A_g - 2(t)(\text{gusset plate thickness} + \tfrac{1}{16}\text{ in.}) \\ &= 2.91\text{ in.}^2 - 2(0.233\text{ in.})(\tfrac{3}{8}\text{ in.} + \tfrac{1}{16}\text{ in.}) \\ &= 2.71\text{ in.}^2 \end{aligned}$$

Because $l = 7$ in. > $H = 3\frac{1}{2}$ in., from AISC *Specification* Table D3.1, Case 6,

$$U = 1 - \frac{\bar{x}}{l}$$

$$\begin{aligned} \bar{x} &= \frac{B^2 + 2BH}{4(B+H)} \quad \text{from AISC } \textit{Specification} \text{ Table D3.1} \\ &= \frac{(3\tfrac{1}{2}\text{ in.})^2 + 2(3\tfrac{1}{2}\text{ in.})(3\tfrac{1}{2}\text{ in.})}{4(3\tfrac{1}{2}\text{ in.} + 3\tfrac{1}{2}\text{ in.})} \\ &= 1.31\text{ in.} \end{aligned}$$

Therefore,

$$\begin{aligned} U &= 1 - \frac{\bar{x}}{l} \\ &= 1 - \frac{1.31\text{ in.}}{7.00\text{ in.}} \\ &= 0.813 \end{aligned}$$

And,

$$\begin{aligned} A_e &= A_n U \qquad \text{(Spec. Eq. D3-1)} \\ &= 2.71\text{ in.}^2(0.813) \\ &= 2.20\text{ in.}^2 \end{aligned}$$

$$\begin{aligned} P_n &= F_u A_e \qquad \text{(Spec. Eq. D2-2)} \\ &= 58\text{ ksi}(2.20\text{ in.}^2) \end{aligned}$$

$= 128 \text{ kips}$

Using AISC *Specification* Section D2, the available tensile rupture strength is:

LRFD	ASD
$\phi_t = 0.75$ $\phi_t P_n = 0.75(128 \text{ kips})$ $= 96.0 \text{ kips} > 80.0 \text{ kips}$ **o.k.**	$\Omega_t = 2.00$ $\dfrac{P_n}{\Omega_c} = \dfrac{128 \text{ kips}}{2.00}$ $= 64.0 \text{ kips} > 52.0 \text{ kips}$ **o.k.**

Available Strength of ³⁄₁₆-in. Weld of Plate to HSS

From AISC *Manual* Part 8, the available strength of a ³⁄₁₆-in. fillet weld is:

LRFD	ASD
$\phi R_n = 4(1.392Dl)$ (from *Manual* Eq. 8-2a) $= 4(1.392)(3 \text{ sixteenths})(7.00 \text{ in.})$ $= 117 \text{ kips} > 80.0 \text{ kips}$ **o.k.**	$\dfrac{R_n}{\Omega} = 4(0.928Dl)$ (from *Manual* Eq. 8-2b) $= 4(0.928)(3 \text{ sixteenths})(7.00 \text{ in.})$ $= 78.0 \text{ kips} > 52.0 \text{ kips}$ **o.k.**

HSS Shear Rupture Strength at Welds (weld on one side)

$$t_{min} = \frac{3.09D}{F_u} \qquad \text{(\textit{Manual} Eq. 9-2)}$$

$$= \frac{3.09(3)}{58}$$

$$= 0.160 \text{ in.} < 0.233 \text{ in.} \qquad \textbf{o.k.}$$

Gusset Plate Shear Rupture Strength at Welds (weld on two sides)

$$t_{min} = \frac{6.19D}{F_u} \qquad \text{(\textit{Manual} Eq. 9-3)}$$

$$= \frac{6.19(3)}{58}$$

$$= 0.320 \text{ in.} < \tfrac{3}{8} \text{ in.} \qquad \textbf{o.k.}$$

A complete check of the connection would also require consideration of the limit states of the other connection elements, such as:

- Whitmore buckling
- Local capacity of column web yielding and crippling
- Yielding of gusset plate at gusset-to-column intersection

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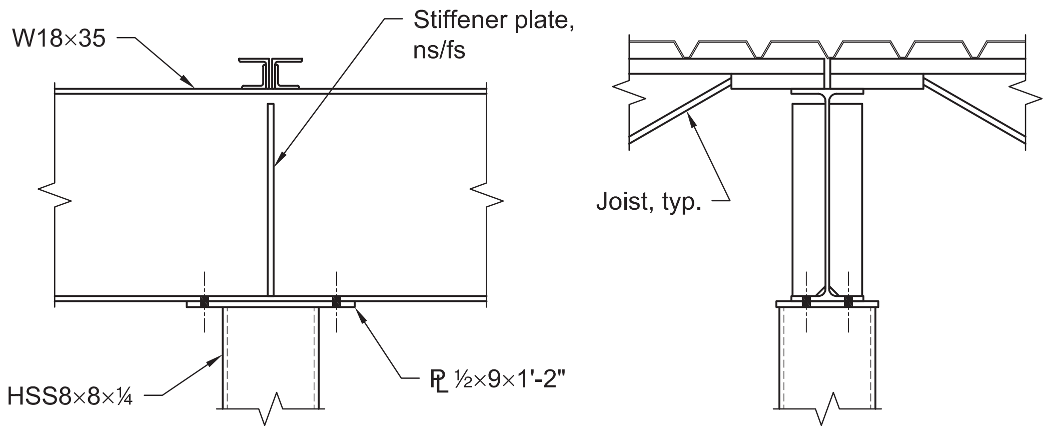
EXAMPLE K.11 RECTANGULAR HSS COLUMN WITH A CAP PLATE, SUPPORTING A CONTINUOUS BEAM

Given:

Verify the local strength of the ASTM A500 Grade B HSS8×8×¼ column subject to the given ASTM A992 W18×35 beam reactions through the ASTM A36 cap plate. Out of plane stability of the column top is provided by the beam web stiffeners; however, the stiffeners will be neglected in the column strength calculations. The column axial forces are R_D = 24 kips and R_L = 30 kips.



Solution:

From AISC *Manual* Tables 2-4 and 2-5, the material properties are as follows:

Beam
ASTM A992
F_y = 50 ksi
F_u = 65 ksi

Column
ASTM A500 Grade B
F_y = 46 ksi
F_u = 58 ksi

Cap Plate
ASTM A36
F_{yp} = 36 ksi
F_u = 58 ksi

From AISC *Manual* Tables 1-1 and 1-12, the geometric properties are as follows:

W18×35
d = 17.7 in.
b_f = 6.00 in.
t_w = 0.300 in.
t_f = 0.425 in.
k_1 = ¾ in.

HSS8×8×¼
t = 0.233 in.

From Chapter 2 of ASCE/SEI 7, the required strength is:

LRFD	ASD
$R_u = 1.2(24.0 \text{ kips}) + 1.6(30.0 \text{ kips})$ $= 76.8 \text{ kips}$	$R_a = 24.0 \text{ kips} + 30.0 \text{ kips}$ $= 54.0 \text{ kips}$

Assume the vertical beam reaction is transmitted to the HSS through bearing of the cap plate at the two column faces perpendicular to the beam.

Bearing Length, l_b, at Bottom of W18×35

Assume the dispersed load width, $l_b = 5t_p + 2k_1$. With $t_p = t_f$.

$$\begin{aligned} l_b &= 5t_f + 2k_1 \\ &= 5(0.425 \text{ in.}) + 2(\tfrac{3}{4} \text{ in.}) \\ &= 3.63 \text{ in.} \end{aligned}$$

Available Strength: Local Yielding of HSS Sidewalls

Determine the applicable equation from AISC *Specification* Table K1.2.

$$\begin{aligned} 5t_p + l_b &= 5(\tfrac{1}{2} \text{ in.}) + 3.63 \text{ in.} \\ &= 6.13 \text{ in.} < 8.00 \text{ in.} \end{aligned}$$

Therefore, only two walls contribute and AISC *Specification* Equation K1-14a applies.

$$\begin{aligned} R_n &= 2F_y t(5t_p + l_b) \\ &= 2(46 \text{ ksi})(0.233 \text{ in.})(6.13 \text{ in.}) \\ &= 131 \text{ kips} \end{aligned}$$ (*Spec*. Eq. K1-14a)

The available wall local yielding strength of the HSS is:

LRFD	ASD
$\phi = 1.00$ $\phi R_n = 1.00(131 \text{ kips})$ $= 131 \text{ kips} > 76.8 \text{ kips}$ **o.k.**	$\Omega = 1.50$ $\dfrac{R_n}{\Omega} = \dfrac{131 \text{ kips}}{1.50}$ $= 87.3 \text{ kips} > 54.0 \text{ kips}$ **o.k.**

Available Strength: Local Crippling of HSS Sidewalls

From AISC *Specification* Table K1.2, the available wall local crippling strength of the HSS is determined as follows:

$$R_n = 1.6t^2\left[1 + \frac{6l_b}{B}\left(\frac{t}{t_p}\right)^{1.5}\right]\sqrt{EF_y\frac{t_p}{t}}$$ (*Spec*. Eq. K1-15)

$$\begin{aligned} &= 1.6(0.233 \text{ in.})^2\left[1 + \left(\frac{6(6.13 \text{ in.})}{8.00 \text{ in.}}\right)\left(\frac{0.233 \text{ in.}}{\tfrac{1}{2} \text{ in.}}\right)^{1.5}\right]\sqrt{29{,}000 \text{ ksi}(46 \text{ ksi})\frac{\tfrac{1}{2} \text{ in.}}{0.233 \text{ in.}}} \\ &= 362 \text{ kips} \end{aligned}$$

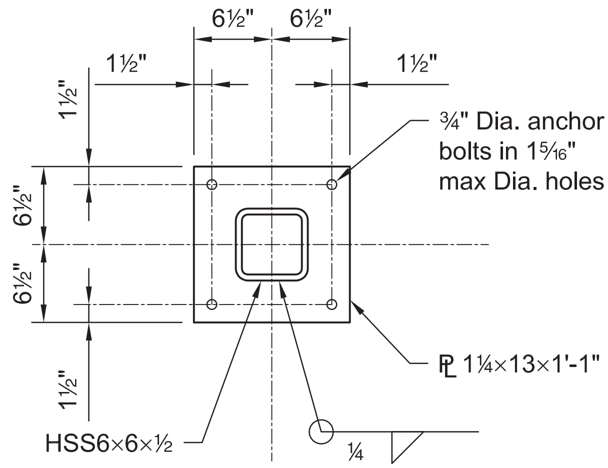
LRFD	ASD
$\phi = 0.75$ $\phi R_n = 0.75(362 \text{ kips})$ $= 272 \text{ kips} > 76.8 \text{ kips}$ **o.k.**	$\Omega = 2.00$ $\dfrac{R_n}{\Omega} = \dfrac{362 \text{ kips}}{2.00}$ $= 181 \text{ kips} > 54.0 \text{ kips}$ **o.k.**

Note: This example illustrates the application of the relevant provisions of Chapter K of the AISC *Specification*. Other limit states should also be checked to complete the design.

EXAMPLE K.12 RECTANGULAR HSS COLUMN BASE PLATE

Given:

An ASTM A500 Grade B HSS6×6×½ column is supporting loads of 40.0 kips of dead load and 120 kips of live load. The column is supported by a 7 ft-6 in. × 7 ft-6 in. concrete spread footing with f_c' = 3,000 psi. Size an ASTM A36 base plate for this column.



Solution:

From AISC *Manual* Tables 2-4 and 2-5, the material properties are as follows:

Column
ASTM A500 Grade B
F_y = 46 ksi
F_u = 58 ksi

Base Plate
ASTM A36
F_y = 36 ksi
F_u = 58 ksi

From AISC *Manual* Table 1-12, the geometric properties are as follows:

HSS6×6×½
$B = H$ = 6.00 in.

From Chapter 2 of ASCE/SEI 7, the required strength is:

LRFD	ASD
P_u = 1.2(40.0 kips) + 1.6(120 kips) = 240 kips	P_a = 40.0 kips + 120 kips = 160 kips

Note: The procedure illustrated here is similar to that presented in AISC Design Guide 1, *Base Plate and Anchor Rod Design* (Fisher and Kloiber, 2006), and Part 14 of the AISC *Manual*.

Try a base plate which extends 3½ in. from each face of the HSS column, or 13 in. × 13 in.

Available Strength for the Limit State of Concrete Crushing

On less than the full area of a concrete support,

$$P_p = 0.85 f_c' A_1 \sqrt{A_2 / A_1} \;\leq\; 1.7 f_c' A_1 \qquad (Spec.\ \text{Eq. J8-2})$$

$$\begin{aligned} A_1 &= BN \\ &= 13.0 \text{ in.}(13.0 \text{ in.}) \\ &= 169 \text{ in.}^2 \end{aligned}$$

$$\begin{aligned} A_2 &= 90.0 \text{ in.}(90.0 \text{ in.}) \\ &= 8{,}100 \text{ in.}^2 \end{aligned}$$

$$\begin{aligned} P_p &= 0.85(3 \text{ ksi})(169 \text{ in.}^2)\sqrt{\frac{8{,}100 \text{ in.}^2}{169 \text{ in.}^2}} \leq 1.7(3 \text{ ksi})(169 \text{ in.}^2) \\ &= 2{,}980 \text{ kips} \leq 862 \text{ kips} \end{aligned}$$

Use P_p = 862 kips.

Note: The limit on the right side of AISC *Specification* Equation J8-2 will control when A_2/A_1 exceeds 4.0.

LRFD	ASD
ϕ_c = 0.65 from AISC *Specification* Section J8 $\phi_c P_p = 0.65(862 \text{ kips})$ $= 560 \text{ kips} > 240 \text{ kips}$ **o.k.**	Ω_c = 2.31 from AISC *Specification* Section J8 $\dfrac{P_p}{\Omega_c} = \dfrac{862 \text{ kips}}{2.31}$ $= 373 \text{ kips} > 160 \text{ kips}$ **o.k.**

Pressure Under Bearing Plate and Required Thickness

For a rectangular HSS, the distance *m* or *n* is determined using 0.95 times the depth and width of the HSS.

$$\begin{aligned} m = n &= \frac{N - 0.95(\text{outside dimension})}{2} \qquad (Manual\ \text{Eq. 14-2}) \\ &= \frac{13.0 \text{ in.} - 0.95(6.00 \text{ in.})}{2} \\ &= 3.65 \text{ in.} \end{aligned}$$

The critical bending moment is the cantilever moment outside the HSS perimeter. Therefore, $m = n = l$.

LRFD	ASD
$f_{pu} = \dfrac{P_u}{A_1}$ $= \dfrac{240 \text{ kips}}{169 \text{ in.}^2}$ $= 1.42 \text{ ksi}$ $M_u = \dfrac{f_{pu} l^2}{2}$ $Z = \dfrac{t_p^2}{4}$	$f_{pa} = \dfrac{P_a}{A_1}$ $= \dfrac{160 \text{ kips}}{169 \text{ in.}^2}$ $= 0.947 \text{ ksi}$ $M_a = \dfrac{f_{pa} l^2}{2}$ $Z = \dfrac{t_p^2}{4}$

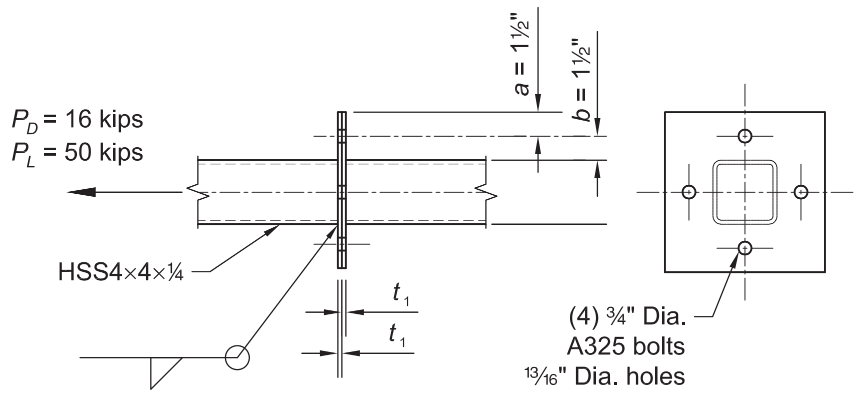
LRFD	ASD
$\phi_b = 0.90$	$\Omega_b = 1.67$
$M_n = M_p = F_y Z$ (*Spec*. Eq. F11-1)	$M_n = M_p = F_y Z$ (*Spec*. Eq. F11-1)
Equating:	Equating:
$M_u = \phi_b M_n$ and solving for t_p gives:	$M_a = M_n/\Omega_b$ and solving for t_p gives:
$t_{p(req)} = \sqrt{\dfrac{2 f_{pu} l^2}{\phi_b F_y}}$	$t_{p(req)} = \sqrt{\dfrac{2 f_{pa} l^2}{F_y / \Omega_b}}$
$= \sqrt{\dfrac{2(1.42\text{ ksi})(3.65\text{ in.})^2}{0.90(36\text{ ksi})}}$	$= \sqrt{\dfrac{2(0.947\text{ ksi})(3.65\text{ in.})^2}{(36\text{ ksi})/1.67}}$
$= 1.08$ in.	$= 1.08$ in.
Or use AISC *Manual* Equation 14-7a:	Or use AISC *Manual* Equation 14-7b:
$t_{p,(req)} = l\sqrt{\dfrac{2P_u}{0.9F_y BN}}$	$t_{p,(req)} = l\sqrt{\dfrac{3.33P_a}{F_y BN}}$
$= 3.65\sqrt{\dfrac{2(240\text{ kips})}{0.9(36\text{ ksi})(13.0\text{ in.})(13.0\text{ in.})}}$	$= 3.65\sqrt{\dfrac{3.33(160\text{ kips})}{(36\text{ ksi})(13.0\text{ in.})(13.0\text{ in.})}}$
$= 1.08$ in.	$= 1.08$ in.

Therefore, use a 1¼ in.-thick base plate.

EXAMPLE K.13 RECTANGULAR HSS STRUT END PLATE

Given:

Determine the weld leg size, end plate thickness, and the size of ASTM A325 bolts required to resist forces of 16 kips from dead load and 50 kips from live load on an ASTM A500 Grade B HSS4×4×¼ section. The end plate is ASTM A36. Use 70-ksi weld electrodes.



Solution:

From AISC *Manual* Tables 2-4 and 2-5, the material properties are as follows:

Strut
ASTM A500 Grade B
F_y = 46 ksi
F_u = 58 ksi

End Plate
ASTM A36
F_y = 36 ksi
F_u = 58 ksi

From AISC *Manual* Table 1-12, the geometric properties are as follows:

HSS4×4×¼
t = 0.233 in.
A = 3.37 in.^{2}

From Chapter 2 of ASCE/SEI 7, the required tensile strength is:

LRFD	ASD
P_u = 1.2(16.0 kips) + 1.6(50.0 kips) = 99.2 kips	P_a = 16.0 kips + 50.0 kips = 66.0 kips

Preliminary Size of the (4) ASTM A325 Bolts

LRFD	ASD
$r_{ut} = \dfrac{P_u}{n}$ $= \dfrac{99.2\text{ kips}}{4}$ $= 24.8$ kips Using AISC *Manual* Table 7-2, try ¾-in.-diameter ASTM A325 bolts. $\phi r_n = 29.8$ kips	$r_{at} = \dfrac{P_a}{n}$ $= \dfrac{66.0\text{ kips}}{4}$ $= 16.5$ kips Using AISC *Manual* Table 7-2, try ¾-in.-diameter ASTM A325 bolts. $\dfrac{r_n}{\Omega} = 19.9$ kips

End-Plate Thickness with Consideration of Prying Action (AISC Manual Part 9)

$$a' = \left(a + \frac{d_b}{2}\right) \le \left(1.25b + \frac{d_b}{2}\right) \qquad \text{(\textit{Manual} Eq. 9-27)}$$

$$= 1.50\text{ in.} + \frac{3/4\text{ in.}}{2} \le 1.25(1.50\text{ in.}) + \frac{3/4\text{ in.}}{2}$$

$$= 1.88\text{ in.} \le 2.25\text{ in.} \qquad \textbf{o.k.}$$

$$b' = b - \frac{d_b}{2} \qquad \text{(\textit{Manual} Eq. 9-21)}$$

$$= 1.50\text{ in.} - \frac{3/4\text{ in.}}{2}$$

$$= 1.13\text{ in.}$$

$$\rho = \frac{b'}{a'} \qquad \text{(\textit{Manual} Eq. 9-26)}$$

$$= \frac{1.13}{1.88}$$

$$= 0.601$$

$d' = 13/16$ in.

The tributary length per bolt,

p = (full plate width)/(number of bolts per side)
= 10.0 in./1
= 10.0 in.

$$\delta = 1 - \frac{d'}{p} \qquad \text{(\textit{Manual} Eq. 9-24)}$$

$$= 1 - \frac{13/16\text{ in.}}{10.0\text{ in.}}$$

$$= 0.919$$

LRFD	ASD
$\beta = \dfrac{1}{\rho}\left(\dfrac{\phi r_n}{r_{ut}} - 1\right)$ (*Manual* Eq. 9-25) $= \dfrac{1}{0.601}\left(\dfrac{29.8 \text{ kips}}{24.8 \text{ kips}} - 1\right)$ $= 0.335$ $\alpha' = \dfrac{1}{\delta}\left(\dfrac{\beta}{1-\beta}\right)$ $= \dfrac{1}{0.919}\left(\dfrac{0.335}{1-0.335}\right)$ $= 0.548 \leq 1.0$	$\beta = \dfrac{1}{\rho}\left(\dfrac{r_n/\Omega}{r_{at}} - 1\right)$ (*Manual* Eq. 9-25) $= \dfrac{1}{0.601}\left(\dfrac{19.9 \text{ kips}}{16.5 \text{ kips}} - 1\right)$ $= 0.343$ $\alpha' = \dfrac{1}{\delta}\left(\dfrac{\beta}{1-\beta}\right)$ $= \dfrac{1}{0.919}\left(\dfrac{0.343}{1-0.343}\right)$ $= 0.568 \leq 1.0$

Use Equation 9-23 for t_{req} in Chapter 9 of the AISC *Manual*, except that F_u is replaced by F_y per recommendation of Willibald, Packer and Puthli (2003) and Packer et al. (2010).

LRFD	ASD
$t_{req} = \sqrt{\dfrac{4 r_{ut} b'}{\phi p F_y (1+\delta\alpha')}}$ $= \sqrt{\dfrac{4(24.8 \text{ kips})(1.13 \text{ in.})}{0.9(10.0 \text{ in.})(36 \text{ ksi})(1+0.919(0.548))}}$ $= 0.480 \text{ in.}$ Use a ½-in. end plate, $t_1 > 0.480$ in., further bolt check for prying not required. Use (4) ¾-in.-diameter A325 bolts.	$t_{req} = \sqrt{\dfrac{\Omega 4 r_{at} b'}{p F_y (1+\delta\alpha')}}$ $= \sqrt{\dfrac{1.67(4)(16.5 \text{ kips})(1.13 \text{ in.})}{10.0 \text{ in.}(36 \text{ ksi})(1+0.919(0.568))}}$ $= 0.477 \text{ in.}$ Use a ½-in. end plate, $t_1 > 0.477$ in., further bolt check for prying not required. Use (4) ¾-in.-diameter A325 bolts.

Required Weld Size

$$R_n = F_{nw} A_{we} \qquad \text{(Spec. Eq. J2-4)}$$

$$\begin{aligned} F_{nw} &= 0.60 F_{EXX}\left(1.0 + 0.50\sin^{1.5}\theta\right) \qquad \text{(Spec. Eq. J2-5)} \\ &= 0.60(70 \text{ ksi})\left(1.0 + 0.50\sin^{1.5} 0.90°\right) \\ &= 63.0 \text{ ksi} \end{aligned}$$

$$\begin{aligned} l &= 4(4.00 \text{ in.}) \\ &= 16.0 \text{ in.} \end{aligned}$$

Note: This weld length is approximate. A more accurate length could be determined by taking into account the curved corners of the HSS.

From AISC *Specification* Table J2.5:

LRFD	ASD
For shear load on fillet welds $\phi = 0.75$ $w \geq \dfrac{P_u}{\phi F_{nw}(0.707)l}$ from AISC *Manual* Part 8 $\geq \dfrac{99.2 \text{ kips}}{0.75(63.0 \text{ ksi})(0.707)(16.0 \text{ in.})}$ $\geq 0.186 \text{ in.}$	For shear load on fillet welds $\Omega = 2.00$ $w \geq \dfrac{\Omega P_a}{F_{nw}(0.707)l}$ from AISC *Manual* Part 8 $\geq \dfrac{2.00(66.0 \text{ kips})}{(63.0 \text{ ksi})(0.707)(16.0 \text{ in.})}$ $\geq 0.185 \text{ in.}$

Try w = 3⁄16 in. > 0.186 in.

Minimum Weld Size Requirements

For t = ¼ in., the minimum weld size = ⅛ in. from AISC *Specification* Table J2.4.

Results:

Use 3⁄16-in. weld with ½-in. end plate and (4) ¾-in.-diameter ASTM A325 bolts, as required for strength in the previous calculation.

CHAPTER K DESIGN EXAMPLE REFERENCES

Fisher, J.M. and Kloiber, L.A. (2006), *Base Plate and Anchor Rod Design,* Design Guide 1, 2nd Ed., AISC, Chicago, IL

Packer, J.A., Sherman, D. and Lecce, M. (2010), *Hollow Structural Section Connections*, Design Guide 24, AISC, Chicago, IL.

Willibald, S., Packer, J.A. and Puthli, R.S. (2003), "Design Recommendations for Bolted Rectangular HSS Flange Plate Connections in Axial Tension," *Engineering Journal*, AISC, Vol. 40, No. 1, 1st Quarter, pp. 15-24.